SMART STAND CONTROL

Complete Source-Accurate Project Reference

Architecture, every gesture family, camera and coordinate logic, face/object following, recording, zoom, training export, platform behavior, exact thresholds, known limitations, test evidence, and source manifest.

Snapshot	Value
Generated	2026-07-23 (Asia/Dhaka)
Branch	followObject
Git HEAD	c4e886dbbfe5
Working tree	dirty - current uncommitted working tree documented
Pinned Flutter	3.41.7 via .fvmrc
App version	1.0.0+1

Prepared directly from the current repository source. Real-hand photographs are credited in Appendix C.

Table of contents

How to read this reference

Accuracy boundary

This document is source-accurate for the current working tree at the snapshot above. It does not claim that a probabilistic ML model will recognize every real hand, that every phone camera behaves identically, or that opaque model weights can be explained from source. Every empirical limitation is separated from deterministic code behavior.

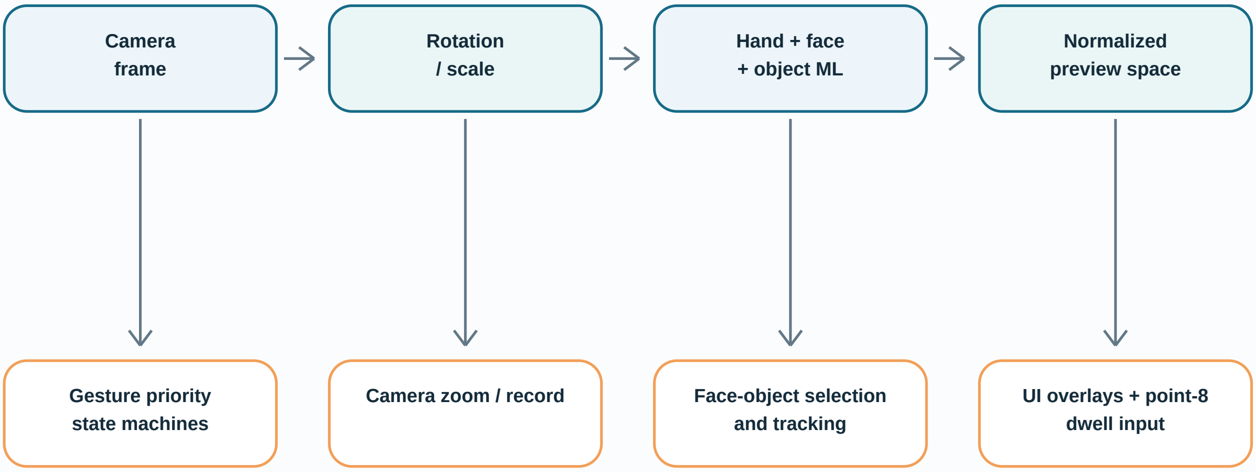
Evidence label	Meaning
Source-proven	Directly read from Dart, Java/Kotlin/Swift, configuration, or tests in this working tree.
Model-dependent	Controlled by TFLite, ML Kit, EfficientDet, YOLO, or device camera internals; source defines inputs/gates but not every learned boundary.
Device-dependent	Varies with lens, sensor orientation, zoom range, camera plugin support, GPU/CPU delegates, storage policy, and OS.
Illustration	Real-hand photographs show pose families only. They are not labeled inference outputs and do not replace exact landmark rules.

Most important project truth

The app detects and displays gestures, changes camera zoom, records silent video, selects visual face/object targets, and updates camera focus/exposure. The repository contains no Bluetooth, serial, network, motor, or stand-controller integration. Therefore Move Left/Right/Up/Down, Stop & Continue, Follow, and Return Main do not physically move a stand.

1. Project purpose and architecture

Smart Stand Control is a Flutter camera application whose primary implemented mode is Hand Gesture. It combines a vendored two-stage hand detector, custom deterministic hand geometry, a canned gesture classifier, ML Kit face detection, one selectable object detector backend, camera recording/zoom, target tracking, and a root point-8 dwell cursor.



App entry	lib/main.dart
Default mode	Hand Gesture
Visible debug entry	Face/Object Debug floating button is enabled
Hidden entry	Moving Down training list item is disabled by a main.dart constant
Home pointer	Enabled; hidden front camera on home, or external live-screen landmarks
Source inventory	363 tracked/untracked text source/config files, 67,646 lines; 95 lib files and 46 test files

1.1 Runtime screen flow

1. main() initializes Flutter, loads the persisted object backend, renders GestureDetectorApp, then starts a non-awaited Ultralytics model prefetch on Android/iOS.
2. The home page offers Automatic Detect, Hand Gesture, and Voice Command. Automatic and Voice only show 'coming soon'.
3. Hand Gesture suspends the hidden home camera, opens AdminHandGestureLiveScreen with the front lens, then resumes the home camera after navigation returns.
4. The debug floating button opens FaceObjectDebugCameraScreen with the selected object backend and front lens.
5. The optional Moving Down page records a two-second raw landmark sample and exports JSONL after review.

1.2 Major code ownership

Area	Primary files	Responsibility
App/home	main.dart; stand_control_home_page.dart; settings_panel.dart	Mode selection, detector preference, navigation, root pointer overlay.
Hand pipeline	hand_detector_factory.dart; third_party/hand_detection	Palm box, 21 landmarks, handedness, world landmarks, canned gestures, ROI tracking.
Gesture domain	custom_gesture_detector.dart; direction_gesture_detector.dart; zoom_gesture_detector.dart; open_palm_gesture_detector.dart	Deterministic landmark geometry and temporal state.
Live orchestration	admin_hand_gesture_live_screen.dart plus five part files	Camera lifecycle, frame loop, priority, recording, zoom, overlays.
Follow	follow_object_sequence_detector.dart; follow_target_selector.dart; target smoother; object_optical_flow_tracker.dart	Sequence, selection, identity, tracking, focus/exposure.
Object ML	five ObjectDetectionService implementations; two Android plugins	Backend-specific preprocessing, inference, validation, common app detections.
Training	moving_down_capture_screen.dart; moving_down_capture_contract.dart; MainActivity.java	Two-second capture, 35-field v2 JSONL review/validation/storage.

2. Camera, coordinates, and hand inference

2.1 Camera ownership and frame cadence

Screen	Preset / audio	Frame format	Special behavior
Home pointer	medium / false	iOS BGRA8888; otherwise YUV420	Front lens preferred; 200/500/1000/2000 ms retries; 2 s watchdog; 5 s stall timeout.
Hand live	high / false	iOS BGRA8888; Android YUV420; other BGRA8888	Portrait UI/capture lock; front/back switch; camera card can rotate portrait/landscape.
Face/object debug	high / false	Same as hand live	Displays raw face/object boxes and exports point 8 to root cursor.
Moving Down	high / false	iOS BGRA8888; otherwise YUV420	Portrait-only two-second collector with review thumbnails.

- Live gesture processing is single-flight: a new frame is ignored while _isProcessing is true.
- Frames are processed no faster than every 50 ms, so the theoretical app-side hand loop ceiling is 20 Hz before inference cost.
- Hand input max dimension is 640. Object-tracking grayscale frames are max 480, or 320 on iOS.
- Camera rotation is derived from raw dimensions, sensor orientation, lens direction, and locked/device orientation. Mirroring is applied separately so detector space remains unmirrored and preview space matches the user.

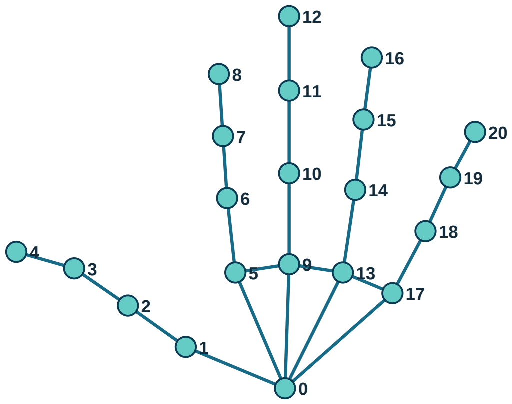
2.2 Hand detector configuration

Pipeline	Palm detection box -> rotated/cropped hand ROI -> full 21-landmark model -> handedness/world landmarks -> optional canned gesture recognition
App configuration	boxesAndLandmarks; full model; palm detector confidence 0.60; max detections 1; minimum landmark score 0.60; tracking on; gestures on
Interpreter policy	iOS forces non-compiled XNNPACK path. Other targets try compiled model first and fall back to XNNPACK on initialization failure.
Tracking defaults	ROI scale 2.0; shift Y -0.1; association IoU 0.5; normalized ROI size 0.03..1.2
Canned classifier	63 image landmark values + 1 handedness value + 63 world values -> 128-D embedder -> 8 probabilities
Classes	Unknown, Closed Fist, Open Palm, Pointing Up, Thumb Down, Thumb Up, Victory, I Love You
Package confidence	Recognizer minimum 0.50; live app also requires package confidence >= 0.50 and reliable hand score >= 0.45

One-hand limit

Although UI state stores a list of hands, HandDetectorFactory configures maxDetections: 1. All primary gesture decisions use the best reliable hand. Multi-hand interaction is therefore not implemented.

2.3 Landmark coordinate map



21-point hand landmark index

0 wrist

1-4 thumb

5-8 index

9-12 middle

13-16 ring

17-20 pinky

Image x: left -> right

Image y: top -> bottom

z: model-relative depth

This diagram is an index map, not a detector output.

ID	Name	High-value uses
0	wrist	Palm centers/plane; direction circle; return-main and punch wrist gate.
1-4	thumb CMC/MCP/IP/tip	Open palm, call-me, OK touch, zoom rays/pinch, fist/tuck checks.
5-8	index MCP/PIP/DIP/tip	Directions; OK; zoom; root point-8 cursor; target debug dwell.
9-12	middle MCP/PIP/DIP/tip	Palm center, folded/open checks; punch center uses 9/10.
13-16	ring MCP/PIP/DIP/tip	Folded/open checks; punch minimum radius anchor 13.
17-20	pinky MCP/PIP/DIP/tip	Palm plane/width, call-me, open-palm spread, folded/open checks.

Geometry convention

Most custom 3D checks weight depth differences by 0.65. A landmark is normally considered visible at visibility >= 0.35; zoom uses 0.30. Hand size is the larger absolute side of the detected hand bounding box. Palm center is the average of visible wrist and four MCP knuckles.

3. Gesture arbitration and UI state

A pose can satisfy several detectors at once. The live screen therefore computes raw custom/package/follow states, builds explicit block reasons, and chooses one user-visible result. Priority is part of correctness, not just presentation.

Priority	Family	What happens
1	No reliable hand / follow release	Continue lost-hand grace or clear follow state. A remembered target can still be refreshed separately.
2	Return-main and face-detect hold	Return-main can clear everything. Call-me can start face selection when no identity is remembered.
3	Follow-object sequence and target selection	Open/fist/release sequence owns gesture flow and suppresses other actions.
4	Recording	OK start, raw Punch pause/resume, Victory stop, each with continuous hold state.
5	Static one-index directions	Left/right/up/down geometry runs if follow/custom overlap/record/punch/package/zoom-transition blockers are absent.
6	Zoom and remaining package labels	Zoom applies only after a direction result is absent and all zoom blockers are clear. Remaining package labels are display feedback.

3.1 Raw custom overlap rules

- CustomGestureDetector can report return-main, OK, call-me, and Punch. Return-main is treated as exclusive cancellation.
- A single custom result may drive recording/call behavior. Multiple simultaneous custom matches create an overlap block and stop direction/zoom fallback.
- Victory has special precedence over pointing directions. Package Pointing Up may directly produce the up direction once steady.
- Punch uses deterministic all-landmark circle geometry. Package Thumb Down is only mapped to the word 'Punch' for display and is not the recording pause gate.
- Follow sequence distinguishes package Closed Fist from compact Punch: if Punch geometry matches, Closed Fist is rejected for the sequence phase.

3.2 Display order

After detection/action updates, visible text is chosen in this order: locked follow target; follow selection failure; no-target release; follow success; remembered-identity unavailable; following hand; active follow sequence; recording feedback; Punch; single custom gesture; custom overlap; zoom result/candidate/hold; movement; known package gesture; generic hand detected.

Action text is not action transport

A status such as 'Moving left', 'Follow the object', or 'Stop & Continue Action' is not evidence of a stand command. There is no command transport layer in pubspec or source.

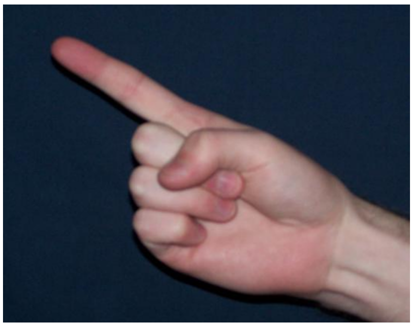
4. Complete 13-gesture catalog

Photograph rule

The real-hand images below are pose-family references selected from Wikimedia Commons. They are not frames from this app, do not show its landmark confidence, and are never used by the detector. Exact source behavior is in each logic block.

4.1 Move Left

Recognizes a static index-pointing-left pose and displays 'Moving left'.



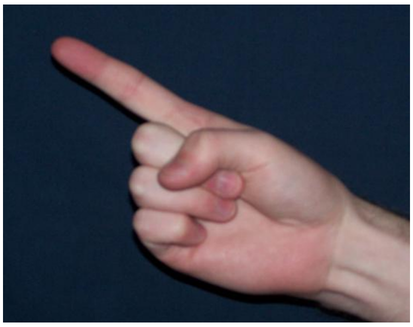
Real hand pose family reference. The app mirrors coordinates so left/right follow the visible preview.

- Keep the hand steady; extend the index finger toward visible-screen left.
- Keep at least one of middle, ring, or pinky folded, or fit the compact palm-circle alternative.
- Do not use a victory sign or a closed pinch.

Exact source logic	• Reliable hand: score >= 0.45 with landmarks. Hand-center movement must stay <= 0.03 hand-size for 3 frames. • Index points 5-6-7-8 define direction. Both 5-6-7 and 6-7-8 joint angles must be >= 145 degrees. • MCP-to-tip direct distance / full 5-6-7-8 path must be >= 0.80. • Direction angle must be 125..235 degrees after preview mirroring. • Index tip must be left of palm center by 0.10..0.15 palm widths; the exact ratio increases when the tip is deeper behind the palm. • Needs 3 consecutive positive frames after the steady-hand gate.
Timing	Frame based: minimum processing interval 50 ms; no time-based swipe window.
Priority	Direction family runs after follow, return-main/call, recording, and custom-overlap blocks. It runs before zoom, but it yields to an armed zoom opening transition.
Implemented effect	UI label only. No motor, Bluetooth, network, or serial stand command exists in the repository.
Limitations	• README wording about a swipe/open-hand movement is not the current implementation. • Fast motion fails the steady-hand gate. Occluded folded fingers can make the pose unavailable. • Only the visible static pose is recognized; distance moved over time is not measured.

4.2 Move Right

Recognizes a static index-pointing-right pose and displays 'Moving right'.



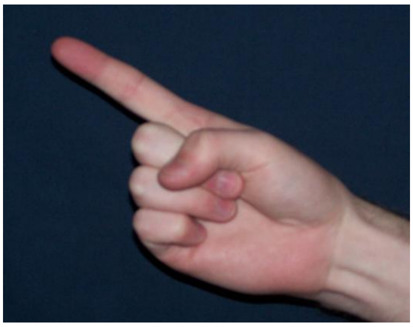
Real hand pose family reference. The photograph points left; direction is determined from live landmark geometry.

- Keep the hand steady and extend the index finger toward visible-screen right.
- Fold at least one of middle, ring, or pinky, or satisfy the compact palm-circle alternative.
- Keep thumb and index clearly separated; a closed pinch is reserved for zoom-out.

Exact source logic	• Reliable-hand and 3-frame steadiness gates match Move Left. • Index straightness must be >= 0.80. Unlike left, the code does not separately require the two 145-degree index joint checks. • Direction angle wraps through zero: >= 305 degrees OR <= 70 degrees. • Index tip must be right of palm center by a depth-aware 0.10..0.15 palm widths. • A zoom-out closed-pinch conflict explicitly rejects the right pose. • Needs 3 consecutive positive frames.
Timing	Frame based; 3 steady frames plus 3 confirming frames can overlap only as implemented by the detector state.
Priority	Same family arbitration as Move Left.
Implemented effect	UI label only; it does not drive physical stand hardware.
Limitations	• Right is intentionally asymmetric with left: no explicit 145-degree two-joint gate in the right branch. • Mirroring errors or an unexpected camera coordinate contract can reverse visible direction. • README swipe wording is outdated.

4.3 Move Up

Recognizes a static upward index pose and displays 'Moving up'.



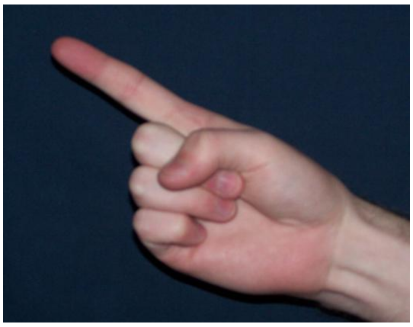
Real hand pose family reference. Rotate the live pose upward; the PDF photo itself is not a detector output.

- Hold points 5, 6, 7, and 8 in strictly rising image order.
- Keep index mostly vertical and at least one other long finger folded.
- Keep the hand center steady for the initial confirmation.

Exact source logic	• 5-8 vertical span must be >= 0.15 palm width; horizontal spread must be <= 0.75 times that span. • Joint 5-6-7 must be >= 135 degrees and 6-7-8 must be >= 170 degrees. • Initial direction sector is 75..120 degrees; while active it widens to 70..125 degrees. • At least one of middle/ring/pinky must be folded, unless the compact palm circle succeeds. • No extra up-only positive-frame counter exists after the shared 3-frame hand-steadiness gate. • A reliable package Pointing Up label can immediately choose up once the hand is steady.
Timing	Static pose; minimum camera processing interval 50 ms.
Priority	Victory package label is rejected before direction. Follow, custom overlaps, recording, punch, and zoom-transition reservations block direction.
Implemented effect	UI label only; no stand elevation command is sent.
Limitations	• The source does not track an upward swipe trajectory. • Because package Pointing Up can select up, package model behavior can differ from geometry-only behavior. • A nearly vertical but slightly reversed landmark chain is rejected.

4.4 Move Down

Recognizes a static downward index pose and displays 'Moving down'.



Real hand pose family reference. Use the same one-index shape with the live finger pointing downward.

- Hold index PIP, DIP, and tip (6-7-8) in strictly descending image order.
- Keep the PIP-to-tip ray mostly vertical and at least one other long finger folded.
- Hold the pose steady until confirmation completes.

Exact source logic	• 6-8 span must be >= 0.15 palm width; horizontal spread <= 0.75 times vertical span. • Joint 6-7-8 must be >= 170 degrees. • Initial sector is 245..295 degrees; active sector is 235..305 degrees. • At least one folded middle/ring/pinky or the compact palm-circle alternative is required. • Needs 3 consecutive positive down frames after steadiness. • The hidden training collector validates downward palm-center travel separately; it is not used by live Move Down recognition.
Timing	3 positive frames, with frames processed no faster than every 50 ms.
Priority	Same direction arbitration as the other three directions.
Implemented effect	UI label only; no motor command is implemented.
Limitations	• Live recognition is not the two-second temporal training logic. • README describes pointing or moving the palm; implementation uses the index chain. • Down is the only vertical direction with a dedicated 3-positive-frame counter.

4.5 Detect My Face

Holds a call-me pose for 2 seconds, runs face detection, and selects a face box.

Real Shaka/call-me hand. The live code evaluates landmark ratios, not image templates.

- Extend thumb and pinky while folding index, middle, and ring.
- Keep thumb and pinky separated and hold continuously for 2 seconds.
- Use only when no follow-target identity is already remembered.

Exact source logic	• Thumb tip distance from 3D palm center must exceed thumb-IP distance * 1.15 and hand-size * 0.23. • Pinky tip must exceed PIP distance * 1.20 and hand-size * 0.30. • Each folded index/middle/ring passes if tip <= PIP distance * 1.03 OR tip < hand-size * 0.26. • Thumb-tip to pinky-tip 3D distance must exceed hand-size * 0.55. • After 2 continuous seconds, the fast ML Kit face detector returns candidates and the presentation layer selects the best face target.
Timing	2-second continuous hold; losing the pose clears the hold.
Priority	Runs before follow-object, recording, direction, and zoom when no remembered target identity blocks it.
Implemented effect	Locks and follows a face bounding box and updates camera focus/exposure. It does not identify a person.
Limitations	• This is face detection, not face recognition, authentication, or identity verification. • A detected face can be missed under profile pose, obstruction, poor light, or scale limits. • The repository contains no consent, biometric storage, or identity database flow.

4.6 Follow The Object

Runs open-palm hold -> closed fist -> final open/relaxed release, then selects and tracks a face or non-person object.

Real open palm used for the first and final phase. Closed-fist phase is shown in the tracking chapter.

- Hold an open palm for 1 second.
- Show the package-classified closed fist. Full-screen face/object candidate scanning then becomes active.
- Move the hand over the target and release with an open palm or at least one relaxed extended long finger for 2 frames.

Exact source logic	• Custom open-palm confidence uses enter 0.55 / exit 0.45 hysteresis, 4 samples, >= 2 positive samples, max age 500 ms. • Closed fist comes from the 8-class package classifier at confidence >= 0.50; a compact Punch-circle match is excluded. • Release point is the center of the hand bounding box, not index point 8. • Candidate memory requires 2 compatible fresh detection cycles and tolerates <= 0.15 normalized hand movement for 2 seconds. • Selection prefers the smallest box containing a point padded by 0.10; the live release path also supports the nearest fresh candidate. • If the hand disappears after fist, a 2-second grace period allows return; timeout auto-releases from the last visible hand center.
Timing	1-second first palm; 2-frame relaxed release; 2-second lost-hand grace; result message 1.2 seconds.
Priority	While active it suppresses custom actions, recording, direction, and zoom. Follow state is the highest long-running activity.
Implemented effect	Selects a box, maintains a target identity record, smooths/predicts the box, and updates camera focus/exposure. It does not move a physical stand.
Limitations	• Object identity is heuristic: tracking ID when available, else type/label/class/spatial continuity and a compact appearance signature. • Occlusion, similar adjacent objects, detector latency, stale held results, and lighting changes can switch or lose a target. • All object backends remove the 'person' class; people are represented only by face boxes.

4.7 Stop and Continue Action

Displays 'Stop & Continue Action' for a reliable package Thumb Up label.

Real thumbs-up reference. Recognition comes from the package classifier, not repo-defined angles.

- Show a thumbs-up pose that the package classifier recognizes.
- Keep the hand detection score and package confidence above their gates.

Exact source logic	• Hand score must be >= 0.45 and package Thumb Up confidence >= 0.50. • The vendored recognizer maps 63 image landmark values + handedness + 63 world values into a 128-D embedding and 8-class classifier. • No local thumbs-up angles, no 1-second timer, and no stop/continue toggle state exist.
Timing	No hold timer in current code, despite README saying 1 second.
Priority	Package labels are considered after higher-priority follow/custom/record/direction/zoom decisions.
Implemented effect	Text feedback only. There is no stop/continue side effect or stand-control command.
Limitations	• The feature name overstates implemented behavior. • Classifier weights and training data are opaque model assets, so exact pose boundaries are not derivable from source. • README timing is inaccurate for the current working tree.

4.8 Return To Main Position

Holds all four long fingers pointing down for 1 second, then clears active gesture tasks and resets camera zoom/focus.

Real open-hand reference. For the live gesture all four long-finger chains must point downward.

- Point index, middle, ring, and pinky downward together.
- Keep each MCP->PIP->DIP->TIP chain descending with visible separation.
- Hold continuously for 1 second.

Exact source logic	• Each long finger needs MCP-PIP-TIP 3D joint angle >= 135 degrees. • Each MCP-to-tip projected distance must be >= 0.20 hand-size. • Every adjacent MCP->PIP->DIP->TIP y-step must descend by >= 0.04 hand-size. • After 1 second, the detector holds the result for 900 ms to avoid flicker. • The live screen clears zoom/direction/follow/record gesture state, target identity/candidates, and optionally resets camera zoom/focus.
Timing	1-second pose hold; 900 ms result latch.
Priority	Return-main is evaluated before face, follow, recording, direction, zoom, and package labels.
Implemented effect	Resets in-app camera and gesture state. It does not physically return a stand to a mechanical home position.
Limitations	• README's circular index-finger description is obsolete and contradicts current source. • All four finger chains must remain visible; partial occlusion breaks the hold. • No hardware homing protocol exists in this repository.

4.9 Start Record Video

Holds a geometry-defined OK sign for 1 second and starts camera video recording.

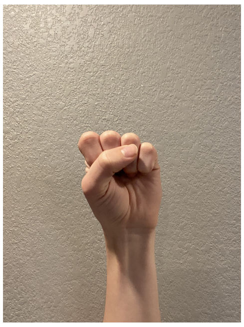
Real OK gesture reference. Exact acceptance uses 3D landmark distances and angles.

- Touch thumb tip to index tip, bend index, and extend middle/ring/pinky.
- Hold for 1 second while not already recording.

Exact source logic	• Thumb-index weighted 3D distance <= max(hand-size * 0.11, 12 pixels). • Index MCP-PIP-tip 3D angle <= 150 degrees. • Middle/ring/pinky must each pass angle-based 3D extension geometry. • At 1 second the stream switches to startVideoRecording(onAvailable: _processCameraImage), with orientation locking and transition overlays. • The camera controller is created with enableAudio: false, so recordings are silent.
Timing	1-second continuous hold; triggers once until the pose changes.
Priority	Recording gestures are disabled during follow. Start requires no conflicting custom overlap.
Implemented effect	Starts a silent camera recording and an elapsed timer.
Limitations	• No audio is recorded even though iOS includes a microphone usage description. • The fixed 12-pixel floor makes the OK touch rule partly scale-dependent. • Camera-plugin recording support varies by device; start errors only show a snackbar and resume streaming best-effort.

4.10 Pause or Resume Video

Holds the custom compact Punch-circle pose for 1 second to toggle recording pause/resume.



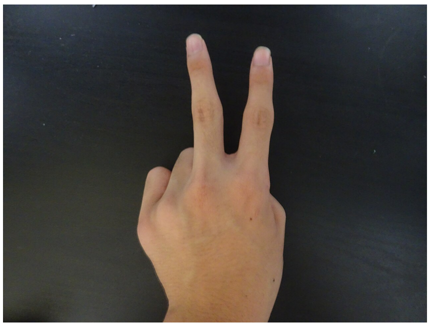
Real closed fist reference. The code calls this recording pose 'Punch' and uses an all-landmarks circle, not the package fist label.

- While recording, compact the complete hand so every landmark fits the defined circle.
- Hold for 1 second; repeat after changing pose to toggle again.

Exact source logic	• Circle center is midpoint of middle MCP/PIP (9/10), or point 10 if point 9 is missing. • Radius = max(0.30 * hand-size, distance between index MCP 5 and ring MCP 13). • All 21 landmarks must be visible and inside; wrist point 0 must be inside. • Normal preview Punch requires 3 steady frames at <= 0.03 hand-size center movement, but recording uses raw one-frame match plus the 1-second hold. • The action calls pauseVideoRecording or resumeVideoRecording and pauses/resumes the app timer.
Timing	1-second continuous hold; only valid while recording.
Priority	Recording action wins over direction and zoom, but follow blocks it.
Implemented effect	Toggles the current silent video recording between paused and active.
Limitations	• README says fist, but package Closed Fist is not the recording gate; exact compact-circle geometry is. • Requiring all 21 reliable landmarks makes the pose sensitive to self-occlusion. • Pause/resume capability depends on the camera plugin and platform implementation.

4.11 End Record Video

Holds a reliable package Victory sign for 2 seconds and stops the recording.



Real victory/peace gesture reference. Classification comes from the vendored model.

- While recording, show a victory sign and keep it continuously recognized for 2 seconds.

Exact source logic	• Hand score >= 0.45 and package Victory confidence >= 0.50. • After 2 seconds, stopVideoRecording returns an XFile; Android then attempts a copy to /storage/emulated/0/Download. • Android filename: smart_stand_recording_<ISO timestamp with ':' and '.' replaced by '-'> plus source extension, default .mp4. • The normal image stream restarts after stop and a short 300 ms transition hold.
Timing	2-second continuous hold; triggers once per unchanged pose.
Priority	Victory blocks direction. The stop mapping only exists while recording; otherwise the UI can show 'Victory'.
Implemented effect	Stops the video and reports it as saved.
Limitations	• On non-Android, _copyRecordingToDownloads returns the original XFile, yet the UI still says 'saved to Download folder'. That message is inaccurate on iOS. • Android copy failures are caught and return the original camera file; the success snackbar can still be misleading. • No gallery/media-library insertion is implemented for recordings.

4.12 Zoom In

Recognizes a stable thumb-index separated pose for 1 second, or immediately continues an armed zoom-out-to-open transition, then adds 0.20 camera zoom.

Real thumb-index pose family reference. For Zoom In the tips must be clearly separated, not touching as in this photograph.

- Fold middle, ring, and pinky; face the palm side toward the camera.
- Place index segment above thumb segment and separate thumb/index tips to >= 0.22 hand-size in both 2D and weighted 3D.
- Hold the palm and at least two folded fingertips stable for 1 second, or open after a completed Zoom Out.

Exact source logic	• Middle/ring/pinky must be folded by 3D angle. Required landmark visibility is 0.30. • Palm-side normalized cross >= 0.10; index segment above thumb by >= 0.02 hand-size. • Thumb/index forward rays must intersect in the handedness-aware screen quadrant, or be parallel within 5 degrees with line separation >= 0.10 hand-size. • Palm movement <= 0.08 hand-size; at least 2 of middle/ring/pinky tips move <= 0.07 hand-size relative to palm. • Static hold is 1 second. Each application changes camera zoom by +0.20 and repeat interval is >= 1 second.
Timing	1-second static hold, except immediate output in the active opening transition; repeat no faster than 1 second.
Priority	Zoom runs only when follow/custom overlap/recording/direction/blocking package/manual zoom are absent. An armed opening transition blocks direction.
Implemented effect	Changes camera zoom level, clamped to device min/max.
Limitations	• The real-image card is only a pose-family cue; exact Zoom In is not the photographed OK sign. • Depth noise, incorrect palm-side orientation, or nearly parallel rays in the wrong quadrant rejects the pose. • A 0.20 level step is not a constant field-of-view percentage across devices.

4.13 Zoom Out

Recognizes a stable closed thumb-index pinch for 1 second and subtracts 0.20 camera zoom; completion arms the opening path for Zoom In.

Real tip-contact reference. Middle, ring, and pinky must be folded in the live Zoom Out pose.

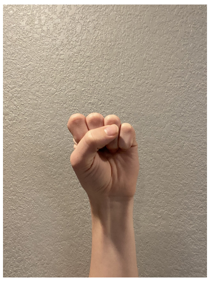
- Fold middle/ring/pinky, face the palm side to camera, keep index segment above thumb, and touch thumb/index tips.
- Hold palm and folded fingertips stable for 1 second.

Exact source logic	• Closed if 2D tip gap <= 0.08 hand-size OR weighted 3D tip gap <= 0.18 hand-size. • A tucked thumb that belongs to a fist/index-point pose rejects the pinch when the tuck check returns false for zoom eligibility. • The same palm-side, visibility, index-above-thumb, and stability gates as Zoom In apply. • After the one-second match, zoom changes by -0.20 and the opening transition becomes armed. • The neutral gap between closed 0.18 and clearly open 0.22 avoids immediate ambiguous static classification.
Timing	1-second static hold; repeat no faster than 1 second.
Priority	Same zoom arbitration. Right-direction logic also explicitly rejects a zoom-out conflict.
Implemented effect	Changes camera zoom level, clamped at the device minimum.
Limitations	• 2D contact can override noisy depth, so perspective overlap may produce a false pinch. • All three other fingers must pass folded-angle checks. • The gesture changes only camera zoom; it has no mechanical stand effect.

5. Face/object selection and tracking

Follow is a layered system: a hand state machine chooses a release point; face/object detectors produce normalized candidates; selection memory confirms one target; identity and spatial gates refresh it; a smoother and frame-to-frame tracker bridge detector gaps; camera focus/exposure follows the target center.

1. Open palm for 1 second



2. Closed fist arms target scan

3. Open or relaxed release

5.1 Follow sequence state machine

Phase	Entry	Exit / failure
idle	No active sequence	Reliable custom open palm starts a one-second hold.
holding first open	Open palm remains continuous	After 1 s -> wait for closed. Interruption resets first hold.
waiting for closed	Sequence remains active with hand visible	Reliable package Closed Fist >= 0.50 -> target selection. Punch-circle fist is excluded.
waiting for final open	Detectors scan candidates and remember hand box center	Custom open palm or >=1 relaxed long finger for 2 frames releases.
waiting for hand return	Hand lost after closed fist; saved center exists	Closed fist can return; release pose completes; 2 s timeout auto-releases saved center.
recent detected latch	Release completed	Displays success for 1.2 s, then returns idle.

5.2 Open-palm scoring

- The custom detector requires a reliable hand and 21-landmark geometry. It rejects visible landmark overlaps closer than 0.012 hand-size and crossed adjacent finger chains.
- Four long-finger extension scores plus thumb, spread, palm-side, image-Y, upper-chain, and adjacent-separation components are combined. Weighted confidence = finger average * 0.60 + spread * 0.15 + palm side * 0.12 + Y * 0.13, then minimum component gates clamp acceptance.
- Finger extension score uses angle 145..165 degrees, tip/PIP ratio 1.08..1.22, and reach 0.25..0.34 hand-size. Thumb score uses angle 125..155 plus reach and index separation.
- Enter threshold is 0.55 and exit threshold 0.45. Four samples younger than 500 ms are retained and at least two must be positive.

5.3 Candidate selection and identity

Stage	Exact behavior	Risk
Candidate pool	Fast ML Kit faces plus non-person objects from selected backend. Detections older than 700 ms are not fresh for selection.	Face and object cycles are asynchronous and can describe different instants.
Point selection	Smallest candidate whose box inflated by 0.10 contains the point; nearest-center fallback is used by live release paths.	Padding can favor a small neighboring box; nearest does not impose a maximum selection distance.
Memory	Requires 2 compatible fresh detector cycles; remembered for 2 s while hand moves <= 0.15 normalized distance.	Ambiguous/missing compatible candidates hide or clear unconfirmed memory.
Identity	Tracking ID if both have one. Else same target type and normalized label; objects also require class index. Faces can compare appearance signature.	Labels/classes are not unique real-world identities.
Spatial continuity	IoU >= 0.08 OR normalized center distance <= 0.18.	A fast jump or long detector gap can break continuity.
Lost state	Two fresh misses are needed; visible box may be held for 900 ms and identity can remain for re-acquisition.	Held boxes are stale estimates, not fresh detections.

Appearance signatures sample the central 80% of a target into an 8x8 representation. They combine an HSV histogram (8 hue x 4 saturation bins), a 64-bit grayscale hash, and aspect ratio. Visible face appearance is accepted at composite similarity >= 0.72. This is a lightweight visual continuity heuristic, not biometric recognition.

5.4 Between-detector tracking

> **Naming versus implementation**
>
> The class/file name says ObjectOpticalFlowTracker, but the current update path performs normalized template correlation with cv.matchTemplate, then transforms/reseeds sparse feature points. It does not call a Lucas-Kanade optical-flow function. The PDF uses 'frame-to-frame template tracker' when describing actual behavior.

- Grayscale input is downscaled to max 480, or 320 on iOS. Android reads the luma plane; iOS can use green as a fast luma proxy.
- Seed uses cv.goodFeaturesToTrack inside the central 80% of the target. It needs at least 12 features, can keep 80, and reseeds below 24 features or after 15 frames.
- Forward and backward template matches use central 80% crops, TM_CCOEFF_NORMED, a search area padded by max(0.75 box size, 0.08), and fixed frame-to-frame scale 1.0.
- Reject if similarity < 0.60, backward error > 1.5, center jump > 0.20, or scale outside 0.70..1.40. Failed tracking releases active mats/points.
- Delayed detector correction compares a historical box by frame ID, applies current delta with blend 0.35, clamps scale ratios 0.70..1.40, and re-seeds without resetting One Euro filters.
- Four One Euro filters smooth box edges (min cutoff 1.0, beta 0.10, derivative cutoff 1.0). Camera focus updates only after >=0.03 movement and >=400 ms.

6. Object detection backends

The home picker persists one backend by enum name. Android defaults to Native YOLO because MethodChannel support is true. iOS defaults to Ultralytics YOLO. Unsupported saved choices fall back to the platform default. Every service is single-flight, so a request arriving during inference is dropped.

Backend	Platform / model	Input and gates	Cadence / stabilization
EfficientDet Lite	All by enum; Android Lite2, iOS Lite0	Android max 640 score >=0.60; iOS max 320 score >=0.35 and deny person; max 5	350 ms, iOS 650 ms; empty hold 800 ms / 3 misses
Ultralytics YOLO	Android/iOS; model id yolo26n; GPU requested	Upright JPEG quality 90; max 640 Android / 416 iOS; confidence 0.45; IoU 0.50; max 5	350/650 ms; empty hold 1200 ms / 3; startup retry 5 s
Google ML Kit	Android/iOS native stream detector	Multiple objects + classification; accepted label >=0.50 else 'Object'; max 5	100 ms Android / 200 ms iOS; empty hold 600 ms / 3
Native YOLO	Android MethodChannel; yolov8n_oiv7.tflite; 601 classes; GPU requested	YUV planes; confidence 0.25; class-aware IoU 0.50; max 5; strict frame/rotation/facing/space validation	250 ms; empty hold 800 ms / 3; startup retry 5 s
OpenCV SDK	Experimental Android Java DNN; yolov8n_oiv7.onnx + TFLite metadata; 601 classes	YUV planes; confidence 0.25; class-aware IoU 0.50; max 5; same response contract validation	400 ms; empty hold 1000 ms / 3; startup retry 5 s

Person ownership

Every AppObjectDetection path filters the exact normalized label 'person'. Faces are supplied by ML Kit FaceDetector instead, preventing a person object box from competing with a face target.

6.1 Preprocessing and coordinate contracts

- Ultralytics converts BGRA/NV21/YUV420 camera data to an upright JPEG in a background isolate. The app maps normalized boxes back into its common image-size contract.
- ML Kit Android accepts direct one-plane NV21 or converts YUV planes to NV21 in background. iOS requires one BGRA8888 plane. Android rotation metadata can swap upright dimensions; iOS native boxes retain raw dimensions because rotation metadata is ignored there.
- Native YOLO and OpenCV send raw planes, row/pixel strides, frame ID, rotation degrees, and camera-facing name across MethodChannel. Results are accepted only when frame ID is new and coordinateSpace == upright_unmirrored with matching rotation/facing.
- All backends sort by confidence when available and cap at 5. ML Kit unclassified boxes have label 'Object', confidence null, class index -1, and may carry native tracking IDs.

6.2 Result stabilization

Each backend has separate empty-result holds, miss limits, normal/fast smoothing alpha, fast-motion threshold, maximum match distance, and partial-track holds. This improves visual continuity but means a visible box can outlive the detector result that created it. Those exact constants are listed in Appendix A.

First-run dependency

Ultralytics model resolution/download is prefetched after runApp and is intentionally not awaited. The UI can start before the model is ready. Network/model metadata failures are absorbed by prefetch, but a later backend startup can still fail and retry.

7. Recording, root hand pointer, debug tools, and training

7.1 Recording lifecycle

Operation	Gesture / UI	Implementation
Start	OK 1 s	Paint loading overlay; wait end-of-frame + 120 ms; lock orientation; stop image stream; startVideoRecording with frame callback; timer starts; keep overlay 300 ms.
Pause/resume	Punch 1 s	Calls camera pause/resume and pauses/resumes elapsed timer; stream callback remains owned by recording API.
Stop	Victory 2 s	Paint overlay; wait; stopVideoRecording; Android copy attempt; unlock to portrait; reset timer; restart normal image stream; keep overlay 300 ms.
Camera switch/dispose	UI/lifecycle	Stops an active recording and attempts save before controller replacement/disposal; best-effort error handling.

Silent recording and save-path mismatch

All camera controllers use enableAudio: false. Android tries /storage/emulated/0/Download. Non-Android returns the camera XFile unchanged, but the snackbar still says 'Recording saved to Download folder'. These are current implementation limitations, not assumptions.

7.2 Root point-8 dwell cursor

- On home, a hidden front camera runs the same hand detector at max dimension 640 and uses the best reliable hand's index tip (landmark 8).
- detectionPointToPreviewCanvas maps detector coordinates through scale, rotation, and front-camera mirroring into the root overlay canvas.
- The overlay performs a Flutter hit test at the cursor and finds the first RenderSemanticsGestureHandler or RenderSemanticsAnnotations with a non-null onTap.
- Holding over the same enabled semantic target for exactly 2 seconds invokes onTap once. Leaving, losing point 8, changing target, or changing owner cancels/restarts progress.
- IgnorePointer makes ordinary touch pass through. Cursor is a yellow 11-pixel circle, black border, green 19-pixel progress arc, and label '8'.
- Before any camera route opens, the home camera is suspended/disposed. Live, debug, and training screens can publish an external pointer without opening a second camera.
- On the normal live gesture screen, showCursor is false when debug mode is off, but the hidden pointer can still activate semantic targets. Tests explicitly verify this behavior.

Operational cost

The home screen continuously owns a camera and runs hand inference while enabled. This can consume battery, heat the device, and conflict briefly with route camera release, which is why retry and watchdog logic exists.

7.3 Hidden gesture diagnostics

A reliable package I Love You pose opens the gesture debug selector after 3 matching frames and latches until 3 release frames. The selector offers one diagnostic family at a time: direction, Punch, Zoom In, Zoom Out, Return Main, recording, Call Me, and Follow Object, plus off/cancel/exit. Point 8 must dwell on a tile for 2 seconds. Opening the selector clears active gesture actions. Normal 21-point overlay remains except while the selector is open.

7.4 Moving Down raw training collector

Item	Exact contract
Availability	Implemented but home list item disabled by showMovingDownTrainingListItem = false.
Capture	Front camera preferred, high preset, no audio, portrait, exactly 2 seconds. Uses hands.first rather than live direction logic.
Record schema	35 fields, schema_version 2. Includes raw dimensions/timestamps, processing FPS, palm/hand metadata, three 21x3 landmark arrays, labels, orientation, camera facing, and aliases.
Review filtering	Only hand_detected == true and complete records remain; no-hand/malformed frames are excluded and valid records are reindexed from zero.
Acceptance	>=12 valid frames; every x/y landmark within inclusive 0.05..0.95; one consistent physical handedness; strongest later palm-center y increase >=0.035.
Palm travel	Average y of points 0,5,9,13,17. Measures largest later increase from highest earlier position, so a slight rise after a valid drop does not erase the sample.
Labels	gesture_target MOVE_DOWN; static label IGNORE_STATIC; temporal label DIRECTION_DOWN; static_loss_enabled false; session_001.
File	userN_direction_down_<UTC compact timestamp>Z.jsonl; one literal JSON object per line with trailing newline.
Storage	Android MediaStore Download/moving down (legacy direct folder pre-Android Q). User IDs scan from user1000. Non-Android uses user1000 and system temp/moving down.
Review images	JPEG max 360, quality 72, aligned to valid records for dialog only; never inserted into JSONL.

Not live classifier training

The collector exports samples only. This repository contains no training script that consumes the JSONL, no newly trained temporal model, and no path that uses this exported data in live Move Down detection.

8. Platform and storage matrix

Capability	Android	iOS	Desktop/web boundary
Primary camera frames	YUV420	One-plane BGRA8888	Live code has BGRA fallback, but main product flags target mobile and object backends vary.
Hand interpreter	Compiled attempted, XNNPACK fallback	Compiled disabled; XNNPACK	Vendored package declares desktop/web plugins, but app workflow is not validated here.
Object backends	All 5; Native YOLO default	EfficientDet, Ultralytics, ML Kit; Ultralytics default	EfficientDet enum is always considered supported; other main flags false.
Face detector	ML Kit fast + tracking	ML Kit fast + tracking	google_mlkit_face_detection is mobile-focused.
Recording	Silent; copy attempt to public Download	Silent; original XFile returned, despite Download snackbar	Not verified.
Training JSONL	MediaStore/direct Download/moving down	System temporary directory	System temporary directory if screen is reachable.
Permissions	CAMERA; WRITE_EXTERNAL_STORAGE only through API 28; camera hardware required	Camera and microphone descriptions; microphone is unused because audio false	Platform-specific permission behavior not documented by app.

README platform drift

README says iOS uses YUV420, but current live/home/debug/training code requests BGRA8888. README also describes audio-capable recordings and several obsolete gestures. Source and passing tests are authoritative for this document.

8.1 Model assets in the repository

Asset	Size	Use
assets/models/yolov8n_oiv7.tflite	14,067,409 bytes	Android Native YOLO inference and OpenCV label metadata
assets/models/yolov8n_oiv7.onnx	14,066,896 bytes	Experimental OpenCV Java DNN
assets/models/yolo26n_w8a32.tflite	2,875,544 bytes	Bundled asset; selected Ultralytics service names model id yolo26n through package resolution
assets/models/yolo26n.mlpackage.zip	2,330,303 bytes	Bundled Apple model archive; package/model preloader controls effective use
hand_detection/hand_detection.tflite	2,339,846 bytes	Palm detector
hand_detection/hand_landmark_full.tflite	5,478,917 bytes	21 image + world landmark model
hand_detection/gesture_embedder.tflite	546,000 bytes	128-D gesture embedding
hand_detection/canned_gesture_classifier.tflite	7,773 bytes	8-class canned gesture probabilities

9. Limitations, mismatches, and failure modes

This chapter consolidates limitations that are otherwise easy to miss when reading individual classes. Severity describes impact on the product claim, not code quality.

Severity	Area	Current limitation	Practical effect
Critical	Stand control	No stand transport or motor API exists.	Movement/follow/home/stop labels do not physically control a stand.
Critical	Product modes	Automatic Detect and Voice Command are placeholders.	Only Hand Gesture opens the primary live feature.
High	Thumb Up	No 1-second hold and no stop/continue state/action.	README and feature name overpromise.
High	Return Main	Current pose is four fingers down for 1 s, not circular index motion.	README/demo instructions are wrong; only in-app state/camera zoom resets.
High	Directions	Static index pose, not swipe/trajectory; output is text only.	Fast swipes can be rejected by steadiness and never command hardware.
High	Recording	enableAudio false; iOS save snackbar says Download while no copy occurs.	Videos are silent and iOS storage messaging is inaccurate.
High	Follow identity	Heuristic labels/classes/tracking IDs/signatures, not persistent real-world identity.	Similar or occluded targets can switch or be lost.
High	Face	Detection only, no recognition/authentication.	'My face' is not verified as the current user.
Medium	Hand count	maxDetections = 1.	No two-hand gestures or robust choice among multiple people.
Medium	ML explainability	Learned model weights/training data are not described in repo.	Exact package pose boundaries and accuracy cannot be guaranteed from code.
Medium	Latency	Single-flight frame drops, backend throttles, asynchronous cycles, stale-result holds.	Visible feedback can lag or briefly show held boxes.
Medium	Tracker	Named optical flow but uses template matching plus feature bookkeeping.	May drift on texture repetition, lighting change, deformation, or occlusion.
Medium	Home camera	Continuously runs on home while enabled.	Battery/thermal/privacy cost and camera route handoff contention.
Medium	Object classes	Person labels are always filtered; backends have different class sets/confidence semantics.	Results differ by backend; unclassified ML Kit objects are generic.
Medium	Storage	Android recording uses direct /storage/emulated/0/Download and catches failures.	Scoped-storage/device policy can leave only the original camera file.
Medium	Training	Collector exports data but no trainer/model integration exists.	Captures do not improve the live detector by themselves.
Low	README	iOS frame format and several gesture descriptions drift from source.	Developers following README can implement/test the wrong pose.
Low	Accessibility	No calibration, handedness preference, hold-duration setting, or alternative input tuning.	Some users/hand shapes/mobility constraints may be underserved.

9.1 Common real-world recognition failures

- Hand too small, cropped, blurred, backlit, motion-blurred, or below hand/landmark confidence gates.
- Occluded fingertips/joints fail visibleLandmark requirements; Punch is especially strict because all 21 points must be inside.
- Front/back mirroring or rotation bugs can reverse a direction or move a box; the project has many mapper tests because this boundary is fragile.
- Depth estimates are model-relative and weighted, not metric camera depth. Perspective and palm orientation can change 3D ratios.
- Frame rate changes convert 'N consecutive frames' into different wall-clock delays. Only hold gestures use explicit DateTime durations.
- Camera zoom ranges and increments are device-specific; a 0.20 zoom-level change is not a universal visual magnification.
- Object detector labels/confidences differ by model. A selection can fail even when a human clearly sees the object.

10. Verification performed for this reference

Check	Result	Meaning / boundary
fvm flutter analyze	PASS - No issues found	Static analysis with pinned Flutter 3.41.7 on the current dirty working tree.
fvm flutter test	PASS - 627 tests	All repository Flutter tests passed. Output ended '+627: All tests passed!'.
Source audit	PASS - app, vendored hand package, native packages, platform entrypoints	Reviewed deterministic logic and generated Appendix A directly from constants; Appendix B hashes source/config files.
Real-hand media	7 source-credited photos	Images are illustrative only and licenses/sources are listed in Appendix C.
Device test	NOT PERFORMED	No physical Android/iOS camera run, performance benchmark, recording storage verification, or real-hand accuracy study in this task.
Coverage	NOT CLAIMED	627 passing tests do not imply every branch, model, plugin, or device behavior is covered.

Branch	followObject
HEAD	c4e886dbbfe5
Working tree	dirty - current uncommitted working tree documented
Changed entries	41
Manifest rows	363
Manifest text lines	67,646

What '100% accurate' can honestly mean

Every deterministic statement in this PDF is tied to the current source snapshot. Runtime recognition accuracy cannot honestly be guaranteed at 100% because it depends on learned models, real hands, lighting, cameras, OS/plugin behavior, and hardware that were not exhaustively tested. Claiming otherwise would be less accurate, not more.

Appendix A. Every HandGestureThresholds constant

The 232 declarations below are parsed directly from lib/hand_gesture_features/domain/constants/hand_gesture_thresholds.dart at generation time. Values are Dart expressions exactly as normalized to one line. This is the exhaustive tuning surface for app-owned gesture, follow, backend, stabilization, tracking, and cadence thresholds.

Type	Constant	Dart value
int	maxDetectionDimension	640
double	minHandScore	0.45
double	minPackageGestureConfidence	0.50
double	openPalmEnterConfidence	0.55
double	openPalmExitConfidence	0.45
double	openPalmMinFingerConfidence	0.50
double	openPalmMinSpreadConfidence	0.50
double	openPalmMinPalmSideConfidence	0.35
double	openPalmMinYAxisConfidence	0.55
double	openPalmMinUpperFingerChainConfidence	0.50
double	openPalmLandmarkOverlapMaxRatio	0.012
double	openPalmAdjacentTipMinDistanceRatio	0.045
double	openPalmAdjacentJointMinDistanceRatio	0.032
double	openPalmMinAdjacentSeparationConfidence	0.50
int	openPalmSmoothingSampleCount	4
int	openPalmSmoothingMinPositiveSamples	2
Duration	openPalmSmoothingMaxAge	Duration(milliseconds: 500)
Duration	followObjectMessageHoldDuration	Duration(milliseconds: 1200,)
Duration	followObjectFirstOpenPalmHoldDuration	Duration(seconds: 1,)
Duration	followObjectHandReturnGraceDuration	Duration(seconds: 2,)
int	followObjectRelaxedReleaseMinExtendedFingers	1
int	followObjectRelaxedReleaseConfirmationFrames	2
int	followTargetSelectionConfirmationCycles	2
Duration	followTargetSelectionMemoryDuration	Duration(seconds: 2,)
double	followTargetSelectionMaxHandMovement	0.15
Duration	followTargetPostReleaseConfirmationDuration	Duration(milliseconds: 1500,)
double	followObjectRelaxedReleaseMinFingerAngleDegrees	145
double	followObjectRelaxedReleaseTipPastPipRatio	1.05
double	followObjectRelaxedReleaseMinReachRatio	0.22
double	followTargetReleasePointPadding	0.10
double	followTargetMinTrackingOverlap	0.08
double	followTargetMaxTrackingDistance	0.18
Duration	followTargetLostHoldDuration	Duration(milliseconds: 900,)
Duration	followTargetDetectionFreshness	Duration(milliseconds: 700,)
Duration	followTargetFreshDetectionWait	Duration(milliseconds: 750,)
int	followTargetLostDetectionCount	2
double	followTargetVisibleSimilarity	0.72
Duration	faceDetectHoldDuration	Duration(seconds: 2)
Duration	objectDetectionMinInterval	Duration(milliseconds: 350,)
Duration	iosObjectDetectionMinInterval	Duration(milliseconds: 650,)

Type	Constant	Dart value
Duration	googleMlKitObjectDetectionMinInterval	Duration(milliseconds: 100,)
Duration	iosGoogleMlKitObjectDetectionMinInterval	Duration(milliseconds: 200,)
Duration	googleMlKitEmptyResultHoldDuration	Duration(milliseconds: 600,)
int	googleMlKitEmptyResultMissLimit	3
double	googleMlKitBoxSmoothingAlpha	0.36
double	googleMlKitFastBoxSmoothingAlpha	0.76
double	googleMlKitFastMotionThreshold	0.06
double	googleMlKitTrackMaxCenterDistance	0.28
double	googleMlKitHandFalsePositivePadding	0.025
double	googleMlKitHandFalsePositiveOverlapRatio	0.45
Duration	googleMlKitPartialTrackHoldDuration	Duration(milliseconds: 400,)
int	googleMlKitPartialTrackMissLimit	3
Duration	nativeMethodChannelObjectDetectionMinInterval	Duration(milliseconds: 250)
Duration	opencvSdkObjectDetectionMinInterval	Duration(milliseconds: 400,)
int	objectDetectionMaxDimension	640
int	objectDetectionMaxResults	5
double	objectDetectionPackageScoreThreshold	0.60
Duration	objectDetectionPackageEmptyResultHoldDuration	Duration(milliseconds: 800)
int	objectDetectionPackageEmptyResultMissLimit	3
double	objectDetectionPackageBoxSmoothingAlpha	0.45
double	objectDetectionPackageFastBoxSmoothingAlpha	0.78
double	objectDetectionPackageFastMotionThreshold	0.08
double	objectDetectionPackageTrackMaxCenterDistance	0.24
Duration	objectDetectionPackagePartialTrackHoldDuration	Duration(milliseconds: 650)
int	objectDetectionPackagePartialTrackMissLimit	2
int	iosObjectDetectionMaxDimension	320
double	iosObjectDetectionPackageScoreThreshold	0.35
String	ultralyticsYoloModelId	'yolo26n'
bool	ultralyticsYoloUseGpu	true
int	ultralyticsYoloMaxDimension	640
int	iosUltralyticsYoloMaxDimension	416
double	ultralyticsYoloConfidenceThreshold	0.45
double	ultralyticsYoloIouThreshold	0.50
int	ultralyticsYoloJpegQuality	90
Duration	ultralyticsYoloEmptyResultHoldDuration	Duration(milliseconds: 1200,)
int	ultralyticsYoloEmptyResultMissLimit	3
double	ultralyticsYoloBoxSmoothingAlpha	0.42
double	ultralyticsYoloFastBoxSmoothingAlpha	0.78
double	ultralyticsYoloFastMotionThreshold	0.08
double	ultralyticsYoloTrackMaxCenterDistance	0.24
Duration	ultralyticsYoloPartialTrackHoldDuration	Duration(milliseconds: 900,)
int	ultralyticsYoloPartialTrackMissLimit	2
Duration	ultralyticsYoloStartupRetryDelay	Duration(seconds: 5)
double	googleMlKitClassificationScoreThreshold	0.50
String	nativeMethodChannelModelAsset	'assets/models/yolov8n_oiv7.tflite'
int	nativeMethodChannelExpectedClassCount	601

Type	Constant	Dart value
bool	nativeMethodChannelUseGpu	true
double	nativeMethodChannelConfidenceThreshold	0.25
double	nativeMethodChannelIouThreshold	0.50
String	opencvSdkModelAsset	'assets/models/yolov8n_oiv7.onnx'
String	opencvSdkMetadataAsset	'assets/models/yolov8n_oiv7.tflite'
int	opencvSdkExpectedClassCount	601
double	opencvSdkConfidenceThreshold	0.25
double	opencvSdkIouThreshold	0.50
Duration	opencvSdkEmptyResultHoldDuration	Duration(milliseconds: 1000,)
int	opencvSdkEmptyResultMissLimit	3
double	opencvSdkBoxSmoothingAlpha	0.42
double	opencvSdkFastBoxSmoothingAlpha	0.78
double	opencvSdkFastMotionThreshold	0.08
double	opencvSdkTrackMaxCenterDistance	0.24
Duration	opencvSdkPartialTrackHoldDuration	Duration(milliseconds: 800,)
int	opencvSdkPartialTrackMissLimit	2
Duration	opencvSdkStartupRetryDelay	Duration(seconds: 5)
Duration	nativeMethodChannelEmptyResultHoldDuration	Duration(milliseconds: 800,)
int	nativeMethodChannelEmptyResultMissLimit	3
double	nativeMethodChannelBoxSmoothingAlpha	0.42
double	nativeMethodChannelFastBoxSmoothingAlpha	0.78
double	nativeMethodChannelFastMotionThreshold	0.08
double	nativeMethodChannelTrackMaxCenterDistance	0.24
Duration	nativeMethodChannelPartialTrackHoldDuration	Duration(milliseconds: 650,)
int	nativeMethodChannelPartialTrackMissLimit	2
Duration	nativeMethodChannelStartupRetryDelay	Duration(seconds: 5,)
int	objectTrackingMaxDimension	480
int	iosObjectTrackingMaxDimension	320
int	objectTrackingHistoryLength	12
int	objectTrackingMaxFeatures	80
int	objectTrackingMinFeatures	12
int	objectTrackingReseedFeatureCount	24
int	objectTrackingReseedFrameCount	15
double	objectTrackingFeatureQuality	0.01
double	objectTrackingFeatureMinDistance	5
double	objectTrackingForwardBackwardError	1.5
double	objectTrackingMinRetention	0.40
double	objectTrackingMinInlierRatio	0.60
double	objectTrackingMaxCenterJump	0.20
double	objectTrackingMinFrameScale	0.70
double	objectTrackingMaxFrameScale	1.40
double	objectTrackingCorrectionBlend	0.35
double	objectTrackingOneEuroMinCutoff	1.0
double	objectTrackingOneEuroBeta	0.10
double	objectTrackingOneEuroDerivativeCutoff	1.0
double	followTargetFocusMovementDeadband	0.03

Type	Constant	Dart value
Duration	followTargetFocusMinInterval	Duration(milliseconds: 400,)
double	minLandmarkVisibility	0.35
double	landmarkDepthWeight	0.65
List<List<HandLandmarkType>>	directionFingerChainTypes	[[HandLandmarkType.indexFingerMCP, HandLandmarkType.indexFingerPIP, HandLandmarkType.indexFingerDIP, HandLandmarkType.indexFingerTip,], [HandLandmarkType.middleFingerMCP, HandLandmarkType.middleFingerPIP, HandLandmarkType.middleFingerDIP, HandLandmarkType.middleFingerTip,], [HandLandmarkType.ringFingerMCP, HandLandmarkType.ringFingerPIP, HandLandmarkType.ringFingerDIP, HandLandmarkType.ringFingerTip,], [HandLandmarkType.pinkyMCP, HandLandmarkType.pinkyPIP, HandLandmarkType.pinkyDIP, HandLandmarkType.pinkyTip,],]
List<HandLandmarkType>	palmReferenceTypes	[HandLandmarkType.indexFingerMCP, HandLandmarkType.middleFingerMCP, HandLandmarkType.ringFingerMCP, HandLandmarkType.pinkyMCP,]
double	directionIndexMinJointAngleDegrees	135.0
double	directionIndexMinProjectedHandSizeRatio	0.20
double	directionSectorHysteresisDegrees	10.0
double	directionMaxHandCenterMovementRatio	0.03
int	directionRequiredSteadyFrames	3
double	verticalDirectionIndexMinAngleDegrees	170.0
double	zoomInMinForwardRayScale	1.0
double	zoomInParallelRayToleranceDegrees	5.0
double	zoomInParallelMinLineSeparationRatio	0.10
double	zoomIndexAboveThumbMinGapRatio	0.02
double	zoomMinPalmSideCross	0.10
double	directionTipMinPalmWidthOffsetRatio	0.10
double	directionTipMaxPalmWidthOffsetRatio	0.15
double	directionTipMaxDepthDeltaPalmWidthRatio	0.25
double	movingLeftIndexMinJointAngleDegrees	145.0
double	movingLeftIndexMinStraightnessRatio	0.80
double	movingLeftMinDirectionAngleDegrees	125.0
double	movingLeftMaxDirectionAngleDegrees	235.0
int	movingLeftRequiredConsecutiveFrames	3
double	movingRightIndexMinStraightnessRatio	0.80
double	movingRightMinDirectionAngleDegrees	305.0
double	movingRightMaxDirectionAngleDegrees	70.0
int	movingRightRequiredConsecutiveFrames	3
double	movingUpMinMcpPipDipJointAngleDegrees	135.0
double	movingUpMinMcpToTipPalmWidthRatio	0.15
double	movingUpMaxHorizontalToVerticalRatio	0.75
double	movingUpInitialMinDirectionAngleDegrees	75.0
double	movingUpInitialMaxDirectionAngleDegrees	120.0
double	movingUpActiveMinDirectionAngleDegrees	70.0
double	movingUpActiveMaxDirectionAngleDegrees	125.0
double	movingDownMinPipToTipPalmWidthRatio	0.15
double	movingDownMaxHorizontalToVerticalRatio	0.75
double	movingDownInitialMinDirectionAngleDegrees	245.0
double	movingDownInitialMaxDirectionAngleDegrees	295.0

Type	Constant	Dart value
double	movingDownActiveMinDirectionAngleDegrees	235.0
double	movingDownActiveMaxDirectionAngleDegrees	305.0
int	directionMinConfirmedFoldedFingerCount	1
List<HandLandmarkType>	directionCompactPalmCircleTypes	[HandLandmarkType.indexFingerMCP, HandLandmarkType.middleFingerMCP, HandLandmarkType.middleFingerPIP, HandLandmarkType.middleFingerDIP, HandLandmarkType.middleFingerTip, HandLandmarkType.ringFingerMCP, HandLandmarkType.ringFingerPIP, HandLandmarkType.ringFingerDIP, HandLandmarkType.ringFingerTip, HandLandmarkType.pinkyMCP, HandLandmarkType.pinkyPIP, HandLandmarkType.pinkyDIP, HandLandmarkType.pinkyTip,]
double	directionCompactPalmCircleRadiusPalmWidthRatio	0.90
double	directionCompactPalmCircleMinImageShortSideRatio	0.15
double	directionFoldedMaxReachAreaRatio	0.7225
double	directionFoldedMaxTopClusterAreaRatio	0.16
double	directionFoldedMaxClusterMcpAreaRatio	0.7225
double	directionFoldedMaxCompressionRatio	0.80
double	directionOpenMinReachAreaRatio	0.81
double	directionOpenMinCompressionRatio	0.85
int	movingDownRequiredConsecutiveFrames	3
double	directionFingerChainMinVerticalImageRatio	0.025
double	directionFingerChainMinVerticalSpanRatio	0.18
double	directionFingerChainVerticalDominanceRatio	1.25
double	directionFingerChainMaxDepthProjectionRatio	1.35
double	fingerTipVerticalMaxSpreadRatio	0.22
double	fingerTipHorizontalMaxSpreadRatio	0.22
double	sideBendMinRatio	0.16
double	rightSideBendMinRatio	0.11
double	rightFingerTipMinOffsetRatio	0.08
int	rightFingerTipMinAlignedCount	3
double	verticalBendMinRatio	0.16
double	wristToMcpAverageMaxRatio	0.30
double	wristToMcpSingleMaxRatio	0.40
double	extendedFingerRatio	1.20
double	foldedFingerRatio	1.03
double	thumbExtendedRatio	1.15
double	fingerFoldedMaxAngleDegrees	145.0
double	fingerExtendedMinAngleDegrees	160.0
double	okTouchMaxDistanceRatio	0.11
double	punchCircleRadiusHandSizeRatio	0.30
List<HandLandmarkType>	punchCircleMinimumRadiusAnchorTypes	[HandLandmarkType.indexFingerMCP, HandLandmarkType.ringFingerMCP,]
int	punchCircleMinInsideLandmarkCount	21
double	punchMaxHandCenterMovementRatio	0.03
int	punchRequiredConsecutiveFrames	3
Duration	returnToMainDownHoldDuration	Duration(seconds: 1)
double	returnToMainFingerMinJointAngleDegrees	135.0
double	returnToMainFingerMinProjectedHandSizeRatio	0.20
double	returnToMainMinAdjacentVerticalGapHandSizeRatio	0.04

Type	Constant	Dart value
Duration	cancelEverythingHoldDuration	Duration(milliseconds: 900,)
double	closedThumbMaxPalmDistanceRatio	0.30
double	closedThumbTipIpPalmRatio	1.00
double	closedThumbMaxKnuckleDistanceRatio	0.32
double	closedThumbMaxTipMcpDistanceRatio	0.36
double	zoomTouchMax2dDistanceRatio	0.08
double	zoomClosedMaxDistanceRatio	0.18
double	zoomInMinDistanceRatio	0.22
double	zoomMaxPalmMovementRatio	0.08
int	zoomStableFingerMinCount	2
double	zoomStableFingerMaxMovementRatio	0.07
Duration	zoomStaticHoldDuration	Duration(seconds: 1)
double	zoomMinLandmarkVisibility	0.30
Duration	gestureZoomRepeatInterval	Duration(seconds: 1)
Duration	minFrameProcessInterval	Duration(milliseconds: 50)
double	zoomStep	0.2
double	gestureZoomStep	0.20
Duration	recordStartHoldDuration	Duration(seconds: 1)
Duration	recordPauseHoldDuration	Duration(seconds: 1)
Duration	recordStopHoldDuration	Duration(seconds: 2)

Appendix B. Source and configuration manifest

This manifest lists current tracked plus untracked text source/config files, excluding tmp, output, .dart_tool, and build. SHA-256 is truncated to 12 hex characters for review. Generated assets and binary model internals are listed elsewhere, not decoded here.

Top-level group	Files
PROJECT_STRUCTURE.md	1
README.md	1
analysis_options.yaml	1
android	13
ios	13
lib	95
packages	29
pubspec.yaml	1
test	46
third_party	163

Path	Lines	SHA-256/12
PROJECT_STRUCTURE.md	152	af1bcfec2f6b
README.md	156	8d7980cf4673
analysis_options.yaml	33	1fb4b7f5e029
android/app/build.gradle.kts	106	17c6e0e16878
android/app/src/debug/AndroidManifest.xml	7	30b0832ed8b9
android/app/src/main/AndroidManifest.xml	49	ae84b1064655
android/app/src/main/java/com/grozziie/gesturedetector/gesture_detector/MainActivity.java	149	1a308515d073
android/app/src/main/res/drawable-v21/launch_background.xml	12	7ca6d5cec216
android/app/src/main/res/drawable/launch_background.xml	12	94433756fe10
android/app/src/main/res/values-night/styles.xml	18	8c5bad5152f2
android/app/src/main/res/values/styles.xml	18	937d7c82f539
android/app/src/profile/AndroidManifest.xml	7	30b0832ed8b9
android/build.gradle.kts	25	c885ddbbdf23
android/gradle.properties	3	20c411ebbe51
android/gradle/wrapper/gradle-wrapper.properties	5	adfec9a4aa85
android/settings.gradle.kts	25	f8ea7fd7a68e
ios/Flutter/AppFrameworkInfo.plist	24	da12038f9b26
ios/Flutter/Debug.xcconfig	2	690b8a8b1ae1
ios/Flutter/Profile.xcconfig	2	39e5af3d86b4
ios/Flutter/Release.xcconfig	2	75abb819a9de
ios/Runner.xcodeproj/project.xcworkspace/xcshareddata/IDEWorkspaceChecks.plist	8	dfa0f9bb85b9
ios/Runner.xcworkspace/xcshareddata/IDEWorkspaceChecks.plist	8	dfa0f9bb85b9
ios/Runner/AppDelegate.swift	16	451c6d9ba0a2
ios/Runner/Assets.xcassets/AppIcon.appiconset/Contents.json	122	3b1e214f466f
ios/Runner/Assets.xcassets/LaunchImage.imageset/Contents.json	23	ccc7950fd5f5
ios/Runner/Assets.xcassets/LaunchImage.imageset/README.md	5	a985b626c4f7
ios/Runner/Info.plist	74	49a895a566f7
ios/Runner/Runner-Bridging-Header.h	1	1a8d11d01d1c

Path	Lines	SHA-256/12
ios/RunnerTests/RunnerTests.swift	12	5e8f4908dace
lib/hand_gesture_features/data/factories/hand_detector_factory.dart	42	39a50ddd0f96
lib/hand_gesture_features/domain/constants/hand_gesture_thresholds.dart	446	48096777e475
lib/hand_gesture_features/domain/enums/camera_preview_mode.dart	10	3d9d7d587c1e
lib/hand_gesture_features/domain/enums/follow_object_release_reason.dart	2	75ae97bf36cf
lib/hand_gesture_features/domain/enums/follow_object_sequence_phase.dart	19	a5bc779c2f0c
lib/hand_gesture_features/domain/enums/follow_target_tracking_phase.dart	2	c49b3c94b6df
lib/hand_gesture_features/domain/enums/follow_target_type.dart	15	57044ce58947
lib/hand_gesture_features/domain/enums/gesture_debug_mode.dart	12	4176e576e320
lib/hand_gesture_features/domain/enums/hand_move_direction.dart	2	61510991cc1a
lib/hand_gesture_features/domain/enums/object_detection_backend.dart	64	1a91c14aed2a
lib/hand_gesture_features/domain/enums/stand_control_mode.dart	2	677e42adf5f5
lib/hand_gesture_features/domain/enums/zoom_direction.dart	2	80c3257163bf
lib/hand_gesture_features/domain/models/app_object_detection.dart	60	aa112f5671a2
lib/hand_gesture_features/domain/models/appearance_signature.dart	57	94baa67dbce0
lib/hand_gesture_features/domain/models/custom_gesture_detection_result.dart	44	d9b8bb9f03a2
lib/hand_gesture_features/domain/models/encoded_yolo_frame.dart	10	73869b8c9bb8
lib/hand_gesture_features/domain/models/follow_object_sequence_result.dart	37	525c808c40e9
lib/hand_gesture_features/domain/models/follow_target.dart	35	9475f5201d18
lib/hand_gesture_features/domain/models/follow_target_identity.dart	46	d7b1c0da727a
lib/hand_gesture_features/domain/models/follow_target_selection_memory.dart	79	0c628ed60754
lib/hand_gesture_features/domain/models/gesture_debug_evaluation.dart	33	58044d9f52b7
lib/hand_gesture_features/domain/models/moving_down_capture_contract.dart	332	0267151d4360
lib/hand_gesture_features/domain/models/object_detection_batch.dart	16	f78c3ac1e943
lib/hand_gesture_features/domain/models/object_optical_flow_track_result.dart	32	cead85a18b79
lib/hand_gesture_features/domain/models/object_tracking_frame.dart	27	4e298e8b7451
lib/hand_gesture_features/domain/models/open_palm_gesture_detection_result.dart	10	794170423254
lib/hand_gesture_features/domain/services/appearance_signature_extractor.dart	260	2341fcba236e
lib/hand_gesture_features/domain/services/custom_gesture_detector.dart	476	5c86582fa7cc
lib/hand_gesture_features/domain/services/direction_gesture_detector.dart	1432	1b92483f36ee
lib/hand_gesture_features/domain/services/follow_object_sequence_detector.dart	553	dfcefea41f0b
lib/hand_gesture_features/domain/services/follow_target_selector.dart	366	e077601cb438
lib/hand_gesture_features/domain/services/follow_target_tracking_progress.dart	34	e8097514c418
lib/hand_gesture_features/domain/services/gesture_debug_evaluator.dart	761	6e5b0f43adaa
lib/hand_gesture_features/domain/services/gesture_debug_menu_trigger.dart	44	33c5e8def101
lib/hand_gesture_features/domain/services/google_mlkit_object_detection_service.dart	292	5df8747e7037
lib/hand_gesture_features/domain/services/hand_geometry_service.dart	1507	4403f3ee1469
lib/hand_gesture_features/domain/services/native_method_channel_object_detection_service.dart	231	792b74b5c945
lib/hand_gesture_features/domain/services/object_detection_backend_preference_service.dart	91	bb488e507be8
lib/hand_gesture_features/domain/services/object_detection_package_service.dart	84	6b112a3af2ba
lib/hand_gesture_features/domain/services/object_detection_request_controller.dart	69	e8daf244869a
lib/hand_gesture_features/domain/services/object_detection_result_stabilizer.dart	94	df6789e7f5d1
lib/hand_gesture_features/domain/services/object_detection_service.dart	79	2234e71a915f
lib/hand_gesture_features/domain/services/object_detection_service_factory.dart	53	6d5b0af96fae
lib/hand_gesture_features/domain/services/object_detection_target_smoother.dart	291	9ae9c2ac546d
lib/hand_gesture_features/domain/services/object_optical_flow_tracker.dart	669	20139df5fb1c

Path	Lines	SHA-256/12
lib/hand_gesture_features/domain/services/open_palm_gesture_detector.dart	839	ec772dbe20f5
lib/hand_gesture_features/domain/services/opencv_sdk_object_detection_service.dart	226	1289661da52b
lib/hand_gesture_features/domain/services/ultralytics_yolo_model_preloader.dart	64	c4e789e05d3b
lib/hand_gesture_features/domain/services/ultralytics_yolo_object_detection_service.dart	182	edaaae1234d5
lib/hand_gesture_features/domain/services/yolo_camera_frame_encoder.dart	206	1aceb3fcf898
lib/hand_gesture_features/domain/services/zoom_gesture_detector.dart	458	47be3c75126f
lib/hand_gesture_features/domain/utils/android_nv21_encoder.dart	57	349065247ca3
lib/hand_gesture_features/domain/utils/camera_frame_box_mapper.dart	168	f55ab554da25
lib/hand_gesture_features/domain/utils/camera_preview_geometry.dart	234	55f37adfc133
lib/hand_gesture_features/domain/utils/detection_debug_log_formatter.dart	19	910079e373e5
lib/hand_gesture_features/domain/utils/moving_down_capture_metadata.dart	29	de19b33a2715
lib/hand_gesture_features/presentation/painters/direction_debug_overlay_painter.dart	1610	32e49ddeb91b
lib/hand_gesture_features/presentation/painters/follow_target_overlay_painter.dart	139	3d3f3fa3b405
lib/hand_gesture_features/presentation/painters/gesture_family_debug_overlay_painter.dart	308	d7377f9e2153
lib/hand_gesture_features/presentation/painters/hand_focus_overlay_painter.dart	181	acbfb13a450f
lib/hand_gesture_features/presentation/painters/hand_landmark_overlay_painter.dart	120	9ac5babeab53
lib/hand_gesture_features/presentation/painters/mobile_stand_painter.dart	134	a067b2c040bd
lib/hand_gesture_features/presentation/painters/object_detection_debug_painter.dart	108	05fbf4d05475
lib/hand_gesture_features/presentation/painters/object_detection_debug_painter_factory.dart	35	f36913a5e29a
lib/hand_gesture_features/presentation/painters/object_optical_flow_debug_painter.dart	79	7472d6a08ac5
lib/hand_gesture_features/presentation/painters/recording_hand_landmark_overlay_painter.dart	200	5605587c36f7
lib/hand_gesture_features/presentation/painters/zoom_in_debug_overlay_painter.dart	473	57121b57b371
lib/hand_gesture_features/presentation/screens/admin_hand_gesture_live_screen.dart	309	b62e8b35a432
lib/hand_gesture_features/presentation/screens/admin_hand_gesture_live_screen_parts/camera_lifecycle.dart	643	f206894b71d1
lib/hand_gesture_features/presentation/screens/admin_hand_gesture_live_screen_parts/gesture_processing.dart	2263	8de8b18c860d
lib/hand_gesture_features/presentation/screens/admin_hand_gesture_live_screen_parts/live_screen_ui.dart	854	bbfdcff1fe00
lib/hand_gesture_features/presentation/screens/admin_hand_gesture_live_screen_parts/recording_controls.dart	785	0eaf75b06bb8
lib/hand_gesture_features/presentation/screens/admin_hand_gesture_live_screen_parts/zoom_controls.dart	381	9ebe34bfbbc1
lib/hand_gesture_features/presentation/screens/face_object_debug_camera_screen.dart	855	3ab055a5877a
lib/hand_gesture_features/presentation/screens/moving_down_capture_screen.dart	730	c78be2fa8bf0
lib/hand_gesture_features/presentation/utils/camera_orientation_preferences.dart	16	d6c2ec068285
lib/hand_gesture_features/presentation/utils/hand_gesture_label_mapper.dart	39	4b7d80d233bb
lib/hand_gesture_features/presentation/utils/ml_kit_preview_mapper.dart	120	12fc9f8bc50a
lib/hand_gesture_features/presentation/utils/palm_orientation_coordinate_policy.dart	12	61d3b38740e4
lib/hand_gesture_features/presentation/widgets/control_mode_card.dart	199	f973742a1c6d
lib/hand_gesture_features/presentation/widgets/gesture_debug_selector_overlay.dart	454	4b7676ab3a21
lib/hand_gesture_features/presentation/widgets/gesture_status_panel.dart	106	1635026dff31
lib/hand_gesture_features/presentation/widgets/hand_camera_loading_view.dart	86	a7322c9f2d48
lib/hand_gesture_features/presentation/widgets/hero_chip.dart	30	d929ca42f252
lib/hand_gesture_features/presentation/widgets/home_hand_pointer_layer.dart	824	58ad1612de5a
lib/hand_gesture_features/presentation/widgets/moving_down_jsonl_review_dialog.dart	270	819309da3de9
lib/hand_gesture_features/presentation/widgets/moving_down_safe_area_overlay.dart	73	dd767dfd8ea1
lib/hand_gesture_features/presentation/widgets/round_icon_button.dart	34	975cfe54d3b4
lib/hand_gesture_features/presentation/widgets/settings_panel.dart	269	51e4ad4ce42a
lib/hand_gesture_features/presentation/widgets/stand_hero_section.dart	91	95d512db1c10
lib/hand_gesture_features/presentation/widgets/touch_zoom_guide_overlay.dart	414	d25f0678fd0b

Path	Lines	SHA-256/12
lib/hand_gesture_features/presentation/widgets/zoom_control_overlay.dart	209	665a499ae859
lib/hand_gesture_features/stand_control_home_page.dart	140	4cef7cc5b2ba
lib/main.dart	261	3ee487dc0692
lib/utils/app_snack_bar.dart	37	0aaad02559e9
packages/native_object_detection/android/build.gradle	53	8ead5eb687c1
packages/native_object_detection/android/src/main/AndroidManifest.xml	1	343d5be1ac17
packages/native_object_detection/android/src/main/kotlin/com/grozziie/native_object_detection/LiteRtRunner.kt	206	40fdb151b577
packages/native_object_detection/android/src/main/kotlin/com/grozziie/native_object_detection/ModelMetadataReader.kt	143	9150793118c7
packages/native_object_detection/android/src/main/kotlin/com/grozziie/native_object_detection/NativeDetectionModels.kt	55	db34be54e846
packages/native_object_detection/android/src/main/kotlin/com/grozziie/native_object_detection/NativeObjectDetectionPlugin.kt	237	2d1b727884cc
packages/native_object_detection/android/src/main/kotlin/com/grozziie/native_object_detection/NativeYoloDetector.kt	87	7f1a9f3e80b6
packages/native_object_detection/android/src/main/kotlin/com/grozziie/native_object_detection/YoloPostprocessor.kt	275	5dcc29511ba2
packages/native_object_detection/android/src/main/kotlin/com/grozziie/native_object_detection/YuvFramePreprocessor.kt	249	3d4e80804170
packages/native_object_detection/android/src/test/kotlin/com/grozziie/native_object_detection/ModelMetadataReaderTest.kt	60	adea414f518c
packages/native_object_detection/android/src/test/kotlin/com/grozziie/native_object_detection/YoloPostprocessorTest.kt	86	62e142c7a7ac
packages/native_object_detection/android/src/test/kotlin/com/grozziie/native_object_detection/YuvFramePreprocessorTest.kt	94	ef641399f81c
packages/native_object_detection/lib/native_object_detection.dart	45	c22aee5cbe2c
packages/native_object_detection/pubspec.yaml	19	bd84cdcecf8f
packages/opencv_object_detection/android/build.gradle	46	f843c2905ca5
packages/opencv_object_detection/android/src/main/AndroidManifest.xml	1	343d5be1ac17
packages/opencv_object_detection/android/src/main/java/com/grozziie/opencv_object_detection/AndroidIoUtils.java	36	c5a332bed7d5
packages/opencv_object_detection/android/src/main/java/com/grozziie/opencv_object_detection/DetectionModels.java	168	e806f559eae7
packages/opencv_object_detection/android/src/main/java/com/grozziie/opencv_object_detection/OpenCvFramePreprocessor.java	219	ce0aead61a59
packages/opencv_object_detection/android/src/main/java/com/grozziie/opencv_object_detection/OpenCvObjectDetectionPlugin.java	352	8792b9b71d8d
packages/opencv_object_detection/android/src/main/java/com/grozziie/opencv_object_detection/OpenCvObjectDetector.java	261	c8cf1c0a8fc8
packages/opencv_object_detection/android/src/main/java/com/grozziie/opencv_object_detection/UltralyticsMetadataReader.java	165	94300b147fd0
packages/opencv_object_detection/android/src/main/java/com/grozziie/opencv_object_detection/YoloPostprocessor.java	271	6c3e86a32185
packages/opencv_object_detection/android/src/test/java/com/grozziie/opencv_object_detection/AndroidIoUtilsTest.java	41	ae3038cf2599
packages/opencv_object_detection/android/src/test/java/com/grozziie/opencv_object_detection/OpenCvFramePreprocessorTest.java	154	ed9e2d970f42
packages/opencv_object_detection/android/src/test/java/com/grozziie/opencv_object_detection/UltralyticsMetadataReaderTest.java	61	a2d54f709f8a
packages/opencv_object_detection/android/src/test/java/com/grozziie/opencv_object_detection/YoloPostprocessorTest.java	142	12313534e5cf
packages/opencv_object_detection/lib/opencv_object_detection.dart	45	8ef256bae4d8
packages/opencv_object_detection/pubspec.yaml	19	56a2931c1e84
pubspec.yaml	78	b58904edfdb1
test/app_object_detection_test.dart	19	eccd0c722fed
test/appearance_signature_extractor_test.dart	165	7f12f8b0ee85
test/camera_frame_box_mapper_test.dart	144	63c482e6b84a
test/camera_orientation_preferences_test.dart	29	f837d055bce8
test/camera_preview_geometry_test.dart	219	11f26999d449
test/custom_gesture_detector_test.dart	1136	eba3090a0c84
test/detection_debug_log_formatter_test.dart	28	2e60e1291be9
test/direction_debug_overlay_painter_test.dart	444	597481f2fe01
test/direction_gesture_detector_test.dart	2714	12b84380a363
test/follow_object_sequence_detector_test.dart	744	993e0964117a

Path	Lines	SHA-256/12
test/follow_target_overlay_painter_test.dart	27	c01625e91b8e
test/follow_target_selection_memory_test.dart	93	13a75bf453cf
test/follow_target_selector_test.dart	550	baa8c593ca1e
test/follow_target_tracking_progress_test.dart	44	b5eefeb33ecf
test/gesture_debug_evaluator_test.dart	58	6e7eb9cff953
test/gesture_debug_menu_trigger_test.dart	63	8c7def4ceead
test/gesture_debug_selector_overlay_test.dart	267	08b1096ed1a7
test/gesture_family_debug_overlay_painter_test.dart	197	dedbf0f4a394
test/gesture_status_panel_test.dart	94	53dd0163e631
test/hand_geometry_service_test.dart	1310	26d1fb4b7bc4
test/hand_gesture_label_mapper_test.dart	10	3d40d44c89b5
test/home_hand_pointer_layer_test.dart	477	bdf8dd3bb0b7
test/main_app_object_detector_test.dart	56	a2eb47596c9c
test/ml_kit_preview_mapper_test.dart	82	0e5c09674bbc
test/moving_down_capture_guide_test.dart	331	50efd946e72a
test/moving_down_jsonl_review_dialog_test.dart	160	36f287c7a12b
test/moving_down_safe_area_overlay_test.dart	49	80e206a6b5f6
test/native_method_channel_object_detection_service_test.dart	265	9036ecfe378f
test/object_detection_backend_preference_service_test.dart	181	da49000be168
test/object_detection_debug_painter_factory_test.dart	68	2cd87569922f
test/object_detection_debug_painter_test.dart	104	b177a2e23fed
test/object_detection_request_controller_test.dart	204	b2890efb0151
test/object_detection_result_stabilizer_test.dart	186	4fe1c70b3149
test/object_detection_service_test.dart	555	6aa5049e8158
test/object_detection_target_smoother_test.dart	224	026f963059ce
test/object_optical_flow_tracker_test.dart	142	b9213d917af2
test/object_tracking_frame_factory_test.dart	112	fe9d6072ec22
test/open_palm_gesture_detector_test.dart	356	0aadbf76e1e0
test/opencv_sdk_object_detection_service_test.dart	264	f2efa76a00d3
test/palm_orientation_coordinate_policy_test.dart	26	6f83c5decbdf
test/touch_zoom_guide_overlay_test.dart	258	2aaa38fa9fab
test/ultralytics_yolo_model_preloader_test.dart	78	21cca26ebd90
test/widget_test.dart	252	1acefcb50218
test/yolo_camera_frame_encoder_test.dart	200	d91a660e4366
test/zoom_gesture_detector_test.dart	1202	bcc862752f90
test/zoom_in_debug_overlay_painter_test.dart	129	db16539710c1
third_party/hand_detection/CHANGELOG.md	195	f45f106e1715
third_party/hand_detection/README.md	453	4e1bc1d7a73a
third_party/hand_detection/analysis_options.yaml	9	968a973f691c
third_party/hand_detection/android/app/src/main/java/io/flutter/plugins/GeneratedPluginRegistrant.java	34	53c48ab6da67
third_party/hand_detection/android/build.gradle	66	9984696d0242
third_party/hand_detection/android/settings.gradle	1	7ef8d5197c0e
third_party/hand_detection/android/src/main/AndroidManifest.xml	3	c3006376668f
third_party/hand_detection/android/src/main/kotlin/com/hugocornellier/hand_detection/HandDetectionPlugin.kt	38	720ae4c24ec1
third_party/hand_detection/android/src/test/kotlin/com/hugocornellier/hand_detection/HandDetectionPluginTest.kt	27	f64671703958
third_party/hand_detection/example/README.md	127	d190495b5ed1

Path	Lines	SHA-256/12
third_party/hand_detection/example/analysis_options.yaml	28	e0ea485cfdbc
third_party/hand_detection/example/android/app/build.gradle.kts	44	56e8c6d55b29
third_party/hand_detection/example/android/app/src/debug/AndroidManifest.xml	7	30b0832ed8b9
third_party/hand_detection/example/android/app/src/main/AndroidManifest.xml	45	097fda8a9283
third_party/hand_detection/example/android/app/src/main/kotlin/com/hugocornellier/hand_detection_example/MainActivity.kt	5	d34808dcc360
third_party/hand_detection/example/android/app/src/main/res/drawable-v21/launch_background.xml	12	7ca6d5cec216
third_party/hand_detection/example/android/app/src/main/res/drawable/launch_background.xml	12	94433756fe10
third_party/hand_detection/example/android/app/src/main/res/values-night/styles.xml	18	8c5bad5152f2
third_party/hand_detection/example/android/app/src/main/res/values/styles.xml	18	937d7c82f539
third_party/hand_detection/example/android/app/src/profile/AndroidManifest.xml	7	30b0832ed8b9
third_party/hand_detection/example/android/build.gradle.kts	33	9713330d73de
third_party/hand_detection/example/android/gradle.properties	3	20c411ebbe51
third_party/hand_detection/example/android/gradle/wrapper/gradle-wrapper.properties	5	4f0df93978f8
third_party/hand_detection/example/android/settings.gradle.kts	26	f822f8653a25
third_party/hand_detection/example/integration_test/accelerator_control_test.dart	82	8113e3e9e8f0
third_party/hand_detection/example/integration_test/hand_detector_benchmark_test.dart	274	bec714181805
third_party/hand_detection/example/integration_test/hand_detector_integration_test.dart	915	b87a7a8bd3d6
third_party/hand_detection/example/integration_test/hand_detector_isolate_benchmark_test.dart	333	e033159b678d
third_party/hand_detection/example/integration_test/hand_video_driver_test.dart	681	3c47a605deb2
third_party/hand_detection/example/ios/Flutter/AppFrameworkInfo.plist	24	da12038f9b26
third_party/hand_detection/example/ios/Flutter/Debug.xcconfig	1	a82d7765b371
third_party/hand_detection/example/ios/Flutter/Release.xcconfig	1	a82d7765b371
third_party/hand_detection/example/ios/Runner.xcodeproj/project.xcworkspace/xcshareddata/IDEWorkspaceChecks.plist	8	dfa0f9bb85b9
third_party/hand_detection/example/ios/Runner.xcworkspace/xcshareddata/IDEWorkspaceChecks.plist	8	dfa0f9bb85b9
third_party/hand_detection/example/ios/Runner/AppDelegate.swift	16	451c6d9ba0a2
third_party/hand_detection/example/ios/Runner/Assets.xcassets/AppIcon.appiconset/Contents.json	122	3b1e214f466f
third_party/hand_detection/example/ios/Runner/Assets.xcassets/LaunchImage.imageset/Contents.json	23	ccc7950fd5f5
third_party/hand_detection/example/ios/Runner/Assets.xcassets/LaunchImage.imageset/README.md	5	a985b626c4f7
third_party/hand_detection/example/ios/Runner/Info.plist	76	fc8d75cea3d3
third_party/hand_detection/example/ios/Runner/Runner-Bridging-Header.h	1	1a8d11d01d1c
third_party/hand_detection/example/ios/RunnerTests/RunnerTests.swift	27	ba14880a29db
third_party/hand_detection/example/lib/main.dart	3019	5a941fb302c6
third_party/hand_detection/example/linux/flutter/generated_plugin_registrant.h	15	00f93bb5b1fa
third_party/hand_detection/example/linux/runner/my_application.h	18	a044da4b485c
third_party/hand_detection/example/macos/Flutter/Flutter-Debug.xcconfig	1	62290ad8e938
third_party/hand_detection/example/macos/Flutter/Flutter-Release.xcconfig	1	62290ad8e938
third_party/hand_detection/example/macos/Flutter/GeneratedPluginRegistrant.swift	18	baf8a2ddef8b
third_party/hand_detection/example/macos/Runner.xcodeproj/project.xcworkspace/xcshareddata/IDEWorkspaceChecks.plist	8	dfa0f9bb85b9
third_party/hand_detection/example/macos/Runner/AppDelegate.swift	13	ce9dbe45a57c
third_party/hand_detection/example/macos/Runner/Assets.xcassets/AppIcon.appiconset/Contents.json	68	bc6ba8105ca6
third_party/hand_detection/example/macos/Runner/Configs/AppInfo.xcconfig	14	6cf0ecbc56fb
third_party/hand_detection/example/macos/Runner/Configs/Debug.xcconfig	2	8067d533ef65
third_party/hand_detection/example/macos/Runner/Configs/Release.xcconfig	10	640cdd686365
third_party/hand_detection/example/macos/Runner/Configs/Warnings.xcconfig	13	0e341a392b16
third_party/hand_detection/example/macos/Runner/Info.plist	36	1e135300a46c
third_party/hand_detection/example/macos/Runner/MainFlutterWindow.swift	15	65c9613c11bc

Path	Lines	SHA-256/12
third_party/hand_detection/example/macos/RunnerTests/RunnerTests.swift	12	6bc180bc77c7
third_party/hand_detection/example/pubspec.yaml	61	f776b293a256
third_party/hand_detection/example/web/manifest.json	35	b5e580e4f15c
third_party/hand_detection/example/windows/flutter/generated_plugin_registrant.h	15	dbc11635d7b1
third_party/hand_detection/example/windows/runner/flutter_window.h	33	38e0fcd3a0b8
third_party/hand_detection/example/windows/runner/resource.h	16	1fa98588b045
third_party/hand_detection/example/windows/runner/utils.h	19	95a7f7dddb83
third_party/hand_detection/example/windows/runner/win32_window.h	102	7dd03746b19d
third_party/hand_detection/example_web/README.md	17	78c7801af535
third_party/hand_detection/example_web/analysis_options.yaml	28	e0ea485cfdbc
third_party/hand_detection/example_web/lib/main.dart	342	220f48d8d732
third_party/hand_detection/example_web/pubspec.yaml	23	f5d3979f8189
third_party/hand_detection/example_web/web/manifest.json	35	78aacf121901
third_party/hand_detection/lib/hand_detection.dart	91	f0e001fa857f
third_party/hand_detection/lib/hand_detection_native.dart	10	860fd24f4ace
third_party/hand_detection/lib/hand_detection_web.dart	13	9daa1d386988
third_party/hand_detection/lib/src/dart_registration.dart	15	cc4d310fb529
third_party/hand_detection/lib/src/exports/opencv_exports.dart	9	48c491b55d0a
third_party/hand_detection/lib/src/exports/opencv_exports_native.dart	1	69d884cf4cb9
third_party/hand_detection/lib/src/exports/opencv_exports_unsupported.dart	1	98bb1de0dff1
third_party/hand_detection/lib/src/exports/opencv_exports_web.dart	1	9643d0697007
third_party/hand_detection/lib/src/hand_detector.dart	818	0331dd6cfe5f
third_party/hand_detection/lib/src/isolate/hand_detector_core.dart	446	3ad1e8b2b5ae
third_party/hand_detection/lib/src/isolate/hand_detector_isolate.dart	91	fe450ad03f75
third_party/hand_detection/lib/src/models/gesture_recognizer.dart	338	bc4b5f0455d2
third_party/hand_detection/lib/src/models/hand_landmark_model.dart	476	59c603209592
third_party/hand_detection/lib/src/models/palm_detector.dart	359	9408c3bf715d
third_party/hand_detection/lib/src/native/hand_native_lib.dart	64	2db082bf7a09
third_party/hand_detection/lib/src/shared/hand_geometry.dart	391	d39360e17c3d
third_party/hand_detection/lib/src/shared/hand_types.dart	549	0676554baaaa
third_party/hand_detection/lib/src/types.dart	6	298617e63c86
third_party/hand_detection/lib/src/ui/hand_overlay.dart	388	854e99ab8f63
third_party/hand_detection/lib/src/util/image_utils.dart	387	4e7b5ae8ae95
third_party/hand_detection/lib/src/web/hand_detector_web.dart	420	a5bc97612286
third_party/hand_detection/lib/src/web/hand_web_lib.dart	60	b10ab7b5ff02
third_party/hand_detection/lib/src/web/models/gesture_recognizer_web.dart	123	abb8e97364fc
third_party/hand_detection/lib/src/web/models/hand_landmark_model_web.dart	166	70e38ff4377c
third_party/hand_detection/lib/src/web/models/palm_detector_web.dart	202	d1f6ec61c576
third_party/hand_detection/linux/flutter/generated_plugin_registrant.h	15	00f93bb5b1fa
third_party/hand_detection/linux/hand_detection_plugin_private.h	5	2fb60128578f
third_party/hand_detection/linux/include/hand_detection/hand_detection_plugin.h	26	c4aa10d0440a
third_party/hand_detection/pubspec.yaml	54	88561d2aeddf
third_party/hand_detection/sample_videos/README.md	18	d70fe4f00365
third_party/hand_detection/test/README.md	192	5268aa00d89c
third_party/hand_detection/test/additional_coverage_test.dart	240	fa63ee4c121b
third_party/hand_detection/test/camera_frame_test.dart	61	7ac7f8545a82

Path	Lines	SHA-256/12
third_party/hand_detection/test/compiled_model_test.dart	152	fea417de7bb4
third_party/hand_detection/test/compiled_pool_bench.dart	78	538c0744966d
third_party/hand_detection/test/config_knobs_test.dart	203	106224549814
third_party/hand_detection/test/decode_failure_test.dart	26	82264219d02f
third_party/hand_detection/test/gesture_recognizer_test.dart	308	e88f17a666df
third_party/hand_detection/test/golden_parity.dart	32	16222c9d95fe
third_party/hand_detection/test/hand_landmark_model_test.dart	286	dc6ef9507752
third_party/hand_detection/test/image_utils_test.dart	671	db8f54d70d1c
third_party/hand_detection/test/isolate_roundtrip_test.dart	38	95fb35609a46
third_party/hand_detection/test/palm_detector_test.dart	661	ebe061de5b6c
third_party/hand_detection/test/roi_from_landmarks_test.dart	298	a577c2123758
third_party/hand_detection/test/roi_mat_test.dart	55	f6bdb7a6a8a8
third_party/hand_detection/test/types_test.dart	862	ee710417dd46
third_party/hand_detection/test/web_compile_test.dart	34	e723f95bec78
third_party/hand_detection/third_party/video_player_avfoundation/CHANGELOG.md	274	5b67439d7376
third_party/hand_detection/third_party/video_player_avfoundation/LOCAL_PATCHES.md	38	b627093a298e
third_party/hand_detection/third_party/video_player_avfoundation/README.md	15	a300a11adf1d
third_party/hand_detection/third_party/video_player_avfoundation/darwin/RunnerTests/TestClasses.swift	293	5c03e9f8b283
third_party/hand_detection/third_party/video_player_avfoundation/darwin/RunnerTests/VideoPlayerTests.swift	867	c5fa0dcf8493
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Package.swift	46	d7bf9bde0ed0
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/AVAssetTrackUtils.m	44	e0d2447d5416
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPAVFactory.m	163	efdecd8e43fa
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPCADisplayLink.m	84	4cbffda7954b
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPEventBridge.m	122	344c6624040c
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPFrameUpdater.m	19	5b8576f4eb87
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPNativeVideoViewFactory.m	47	51351e3ed6d1
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPTextureBasedVideoPlayer.m	218	cf55a8cbe50b
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPVideoPlayer.m	520	87311d4d46ff
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPVideoPlayerPlugin.m	317	2a8dbd908d5c
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/FVPViewProvider.m	38	5779c7ad5af8
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/AVAssetTrackUtils.h	12	2292561d9be4
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPAVFactory.h	121	c0bad66253b2
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPAssetProvider.h	34	1b3904069830
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPDisplayLink.h	51	4f29556388b5
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPEventBridge.h	22	68785c72dbbf
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPFrameUpdater.h	36	ad748fbd8fac
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPNativeVideoView.h	23	4bf784924a70
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPNativeVideoViewFactory.h	22	312ada8da923

Path	Lines	SHA-256/12
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPTextureBasedVideoPlayer.h	33	7b74d57c11ea
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPTextureBasedVideoPlayer_Test.h	29	3b8e1e10de6b
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPVideoEventListener.h	32	f262ab82d170
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPVideoPlayer.h	47	b58d12762e57
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPVideoPlayerPlugin.h	17	f4bf7e0816b9
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPVideoPlayerPlugin_Test.h	44	4c4d34c3b2dc
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPVideoPlayer_Internal.h	40	380f62a98f34
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/FVPViewProvider.h	35	068490751cdc
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/include/video_player_avfoundation/messages.g.h	112	ee4a6561c35c
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation/messages.g.m	594	71dc90d5d43c
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation_ios/FVPNativeVideoView.m	38	c2a2575539af
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation_ios/include/FVPEmpty.h	6	53c2eabeba99
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation_macos/FVPCoreVideoDisplayLink.m	99	d7044f07118d
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation_macos/FVPNativeVideoView.m	25	8619f76e1932
third_party/hand_detection/third_party/video_player_avfoundation/darwin/video_player_avfoundation/Sources/video_player_avfoundation_macos/include/FVPEmpty.h	6	53c2eabeba99
third_party/hand_detection/third_party/video_player_avfoundation/lib/src/avfoundation_video_player.dart	379	d675c51ecb7e
third_party/hand_detection/third_party/video_player_avfoundation/lib/src/messages.g.dart	679	f325d1d25c97
third_party/hand_detection/third_party/video_player_avfoundation/lib/video_player_avfoundation.dart	5	d1f6b8ac1989
third_party/hand_detection/third_party/video_player_avfoundation/pigeons/messages.dart	97	0427af501b2c
third_party/hand_detection/third_party/video_player_avfoundation/pubspec.yaml	38	073c8a4de812
third_party/hand_detection/windows/hand_detection_plugin.h	29	f2014f7450a9
third_party/hand_detection/windows/include/hand_detection/hand_detection_plugin.h	15	4e6173aa65ee
third_party/hand_detection/windows/include/hand_detection/hand_detection_plugin_c_api.h	23	e2e3c0e34308

Appendix C. Real-hand image credits

All photographs are loaded from local temporary copies at PDF generation time and reproduced without cropping or rotation; ReportLab only scales them to fit. Each image remains under its stated license. The PDF uses each only as an illustrative pose-family reference.

Use	Wikimedia file / author	License	Source URL
Open palm	Hand.JPG / Jeremie63	Public domain	https://commons.wikimedia.org/wiki/File:Hand.JPG
Pointing index	Index finger 2.JPG / Than217	Public domain	https://commons.wikimedia.org/wiki/File:Index_finger_2.JPG
Closed fist	Closed fist.jpg / Cedar Tree	CC BY-SA 4.0	https://commons.wikimedia.org/wiki/File:Closed_fist.jpg
OK	OK Hand Gesture (cropped).jpg / faceofwiki	CC0	https://commons.wikimedia.org/wiki/File:OK_Hand_Gesture_(cropped).jpg
Call-me/Shaka	Shaka-sign.png / Roblespepe	CC BY-SA 4.0	https://commons.wikimedia.org/wiki/File:Shaka-sign.png
Thumbs up	Thumbs Up.JPG / faceofwiki	CC0	https://commons.wikimedia.org/wiki/File:Thumbs_Up.JPG
Victory	A Peace sign photo.jpg / Moyashi-otaku	CC BY 4.0	https://commons.wikimedia.org/wiki/File:A_Peace_sign_photo.jpg

License notice

Attribution and license names are supplied for each included image. Follow the linked file description for complete license terms. Inclusion does not imply that any photographer endorses this application or document.

Appendix C.1 Primary local sources

- README.md and PROJECT_STRUCTURE.md were used as intent/history references, then checked against source.
- lib/main.dart and lib/hand_gesture_features define the current app behavior.
- third_party/hand_detection is vendored source version 3.3.0 and is included in the audit boundary.
- packages/native_object_detection and packages/opencv_object_detection define the Android app-owned inference bridges.
- android/app and ios/Runner define permissions, storage channel, labels, and platform launch behavior.
- test contains the 627 passing Flutter tests used for behavioral verification.